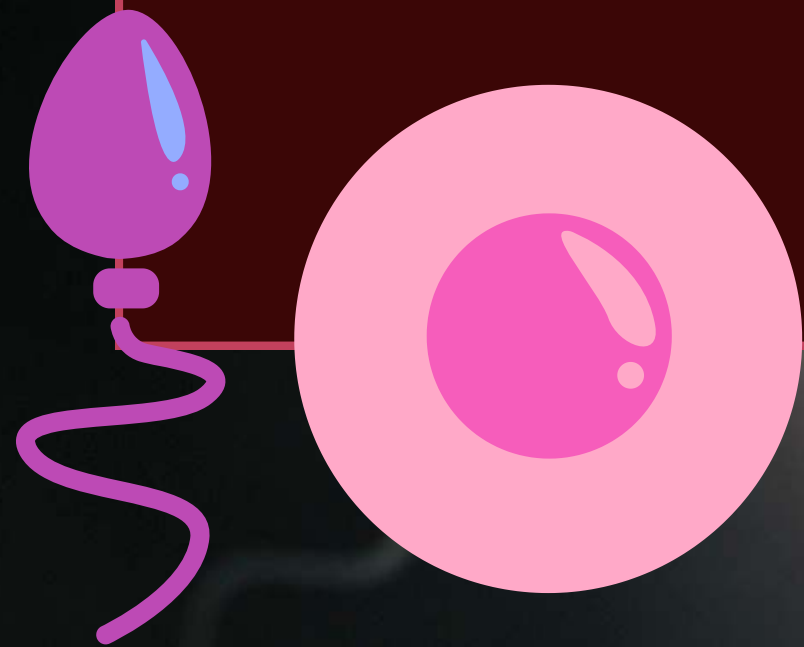


HOW DO ORGANISMS REPRODUCE?



REPRODUCTION:

Reproduction: is a biological process by which living organisms produces new individuals similar to themselves. It ensures continuity of life on earth and helps in evolution of species.

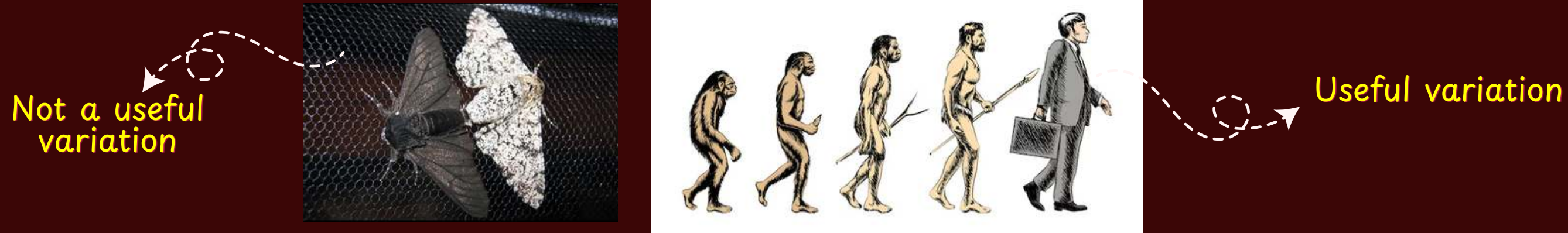
- **DNA (Deoxyribonucleic Acid):** A molecule that carries genetic information in organisms, determining inherited traits and essential for growth, development, and reproduction.
- **Chromatin:** A genetic material composed of DNA, RNA, and proteins, which condenses to form chromosomes in the nucleus of a eukaryotic cell.
- **Chromosomes:** Thread-like structures made of DNA and proteins that carry genetic information from one generation to the next.
- **Genes:** A segment of DNA that contains instructions for a specific trait in an organism.

Advantages of Reproduction:

- Reproduction ensures continuity of a particular species on earth.
- It creates genetic variation which leads to stability of a species.
- It helps in evolution of a species.

VARIATION:

The differences in traits among individuals of a species caused by genetic changes. Variations play a crucial role in adaptation and evolution



Importance:

- **Survival in Changing Environment:** Variations help species adapt to environmental changes, increasing their chances of survival.
- **Basis for Evolution:** Beneficial variations accumulate over generations, leading to the evolution of new species.
- **Disease Resistance:** Some variations provide resistance to diseases, ensuring the survival of certain individuals.
- **Better Adaptation:** Variations help organisms adjust to different habitats and conditions.
- **Continuity of Species:** Variations prevent species from becoming extinct due to sudden environmental changes.

Although variation helps the species, some changes might not be good for an individual in their specific environment, which could make it harder for them to survive.

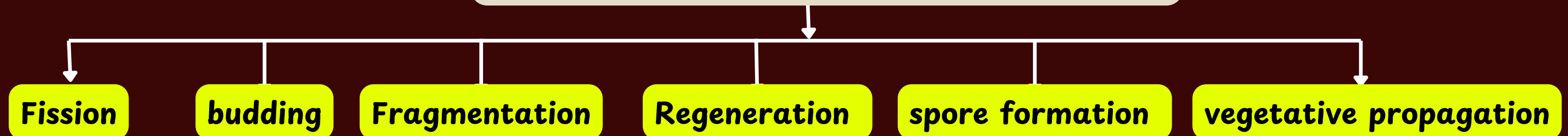
Q. State the importance of variation in species and why it is important.

Answer.

- 1. Survival in Changing Environment: Variations help species adapt to environmental changes, increasing their chances of survival.**
- 2. Basis for Evolution: Beneficial variations accumulate over generations, leading to the evolution of new species.**
- 3. Disease Resistance: Some variations provide resistance to diseases, ensuring the survival of certain individuals.**
- 4. Better Adaptation: Variations help organisms adjust to different habitats and conditions.**
- 5. Continuity of Species: Variations prevent species from becoming extinct due to sudden environmental changes.**

SEXUAL REPRODUCTION	ASEXUAL REPRODUCTION
Two parents are involved	Only one parent is involved
Gamete formation occurs	No gamete formation occurs
Fertilization occurs	Fertilization does not occur
Offsprings formed are genetically diverse	Offsprings are genetically identical to the parent

MODES OF ASEYUAL REPRODUCTION



Q.Reason for the greater similarities among the offsprings produced by asexual reproduction, is:

- (i) Asexual reproduction involves only one parent**
 - (ii) Asexual reproduction involves two parents**
 - (iii) Asexual reproduction involves gametes**
 - (iv) Asexual reproduction does not involve gametes**
-
- (a) (i) and (ii)**
 - (b) (i) and (iii)**
 - (c) (ii) and (iv)**
 - (d) (i) and (iv)**

Q. Write one main difference between asexual and sexual mode of reproduction. Which species is likely to have comparatively better chances of survival – the one reproducing asexually or the one reproducing sexually? Give reason to justify your answer

Answer: Difference between asexual and sexual mode of reproduction is as follows : Asexual reproduction: Gametes are not formed hence fertilisation does not take place. Sexual reproduction: Gametes are always formed and fertilisation takes place to form a zygote. Species reproducing sexually has a better chance of survival as variations occur only during the sexual reproduction. Variations increase the chances of survival of an individual by making them more fit. Selection of variations by environmental factors forms the basis of evolution

Q. Why does asexual reproduction produce genetically identical offspring?

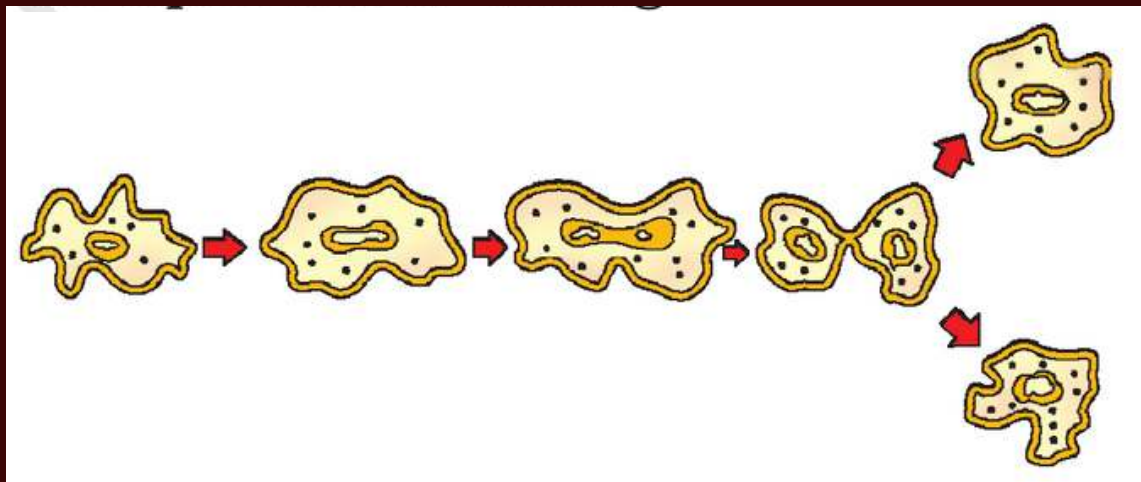
Q. Why does asexual reproduction produce genetically identical offspring?

Answer. In asexual reproduction, offspring are produced through mitosis, which results in the exact copying of parental DNA. Since there is no gamete fusion or genetic recombination, all offspring are clones of the parent.

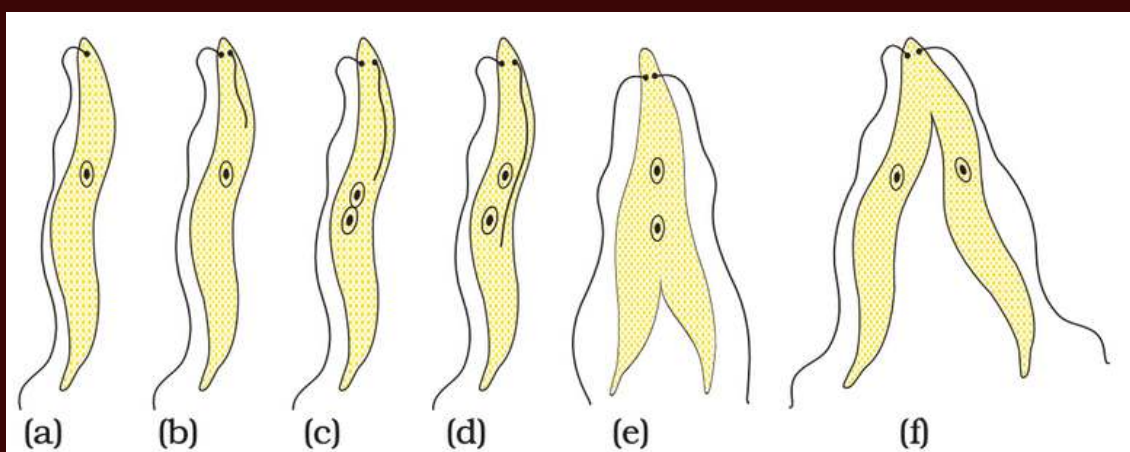
1. Fission (Unicellular Organisms)

Fission: Fission: A type of asexual reproduction in which a unicellular organism splits to form two or more new individuals.

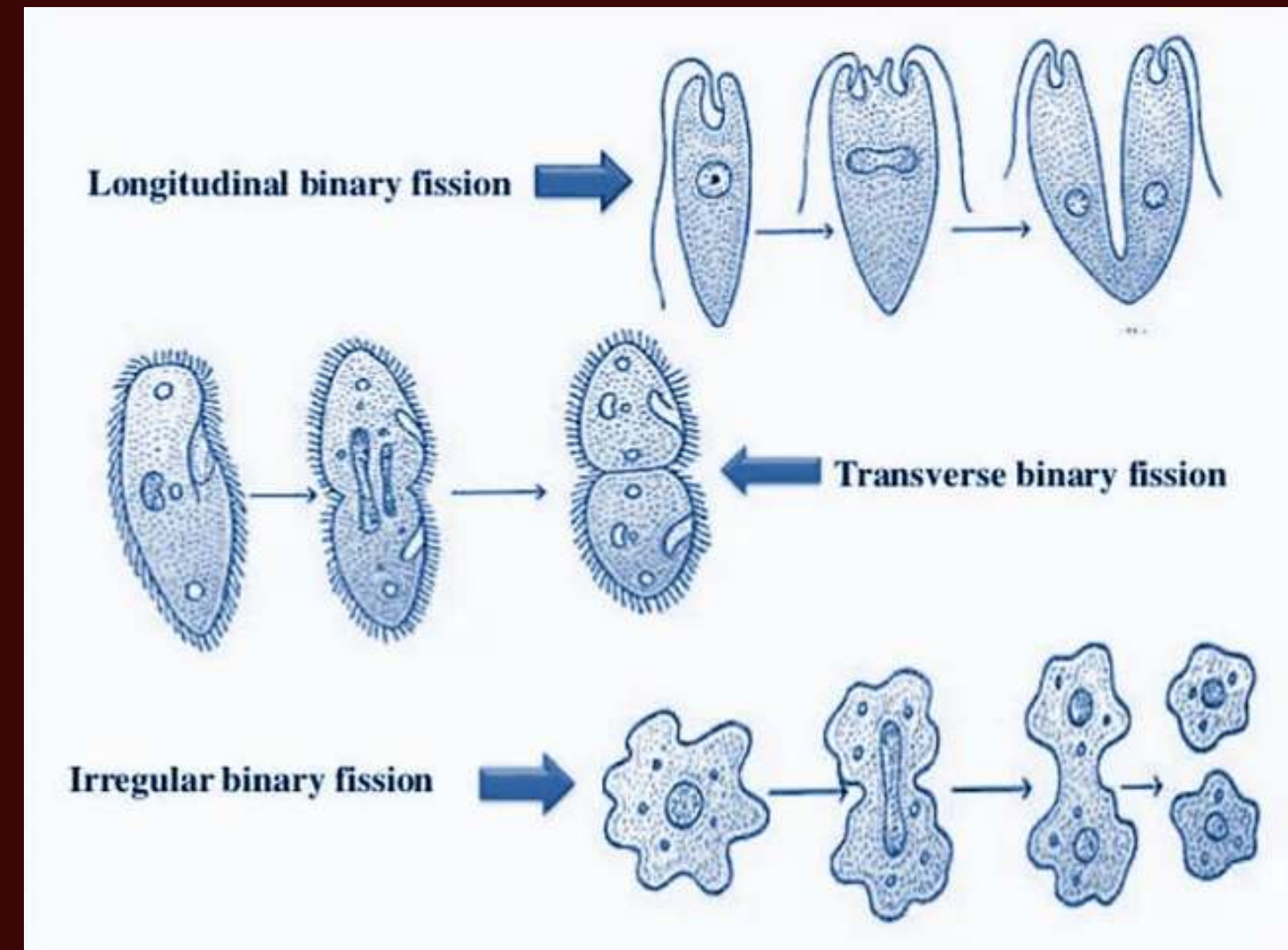
1. Binary Fission: A type of asexual reproduction where a single parent organism divides into two nearly equal-sized daughter cells. The process begins with nuclear division (karyokinesis), followed by cytoplasmic division (cytokinesis), resulting in two identical daughter organisms. Leishmania, a unicellular organism, undergoes binary fission in a specific direction due to its whip-like structure. It causes **Kala-Azar disease**.



Binary fission in Amoeba



Binary fission in Leishmania



2. Multiple Fission: A type of asexual reproduction where the parent organism divides into many small daughter cells. The nucleus divides multiple times before the cytoplasm splits, forming several daughter nuclei. In Plasmodium (malarial parasite), during unfavorable conditions, a tough cyst forms around the cell. When favorable conditions return, the cyst wall breaks, releasing multiple daughter cells.



Binary Fission	Multiple Fission
A single parent cell divides into two daughter cells.	A single parent cell divides into multiple daughter cells at once.
Can be simple (Amoeba) or oriented (Leishmania, divides in a specific direction due to its whip-like structure).	Occurs as a survival mechanism, especially in unfavorable conditions.
Produces two daughter cells.	Produces many daughter cells.
Example: Amoeba, Leishmania.	Example: Plasmodium (malarial parasite).
Occurs in favorable conditions.	Can occur in unfavorable conditions by forming a protective cyst.

2. Fragmentation (Multicellular Organisms)

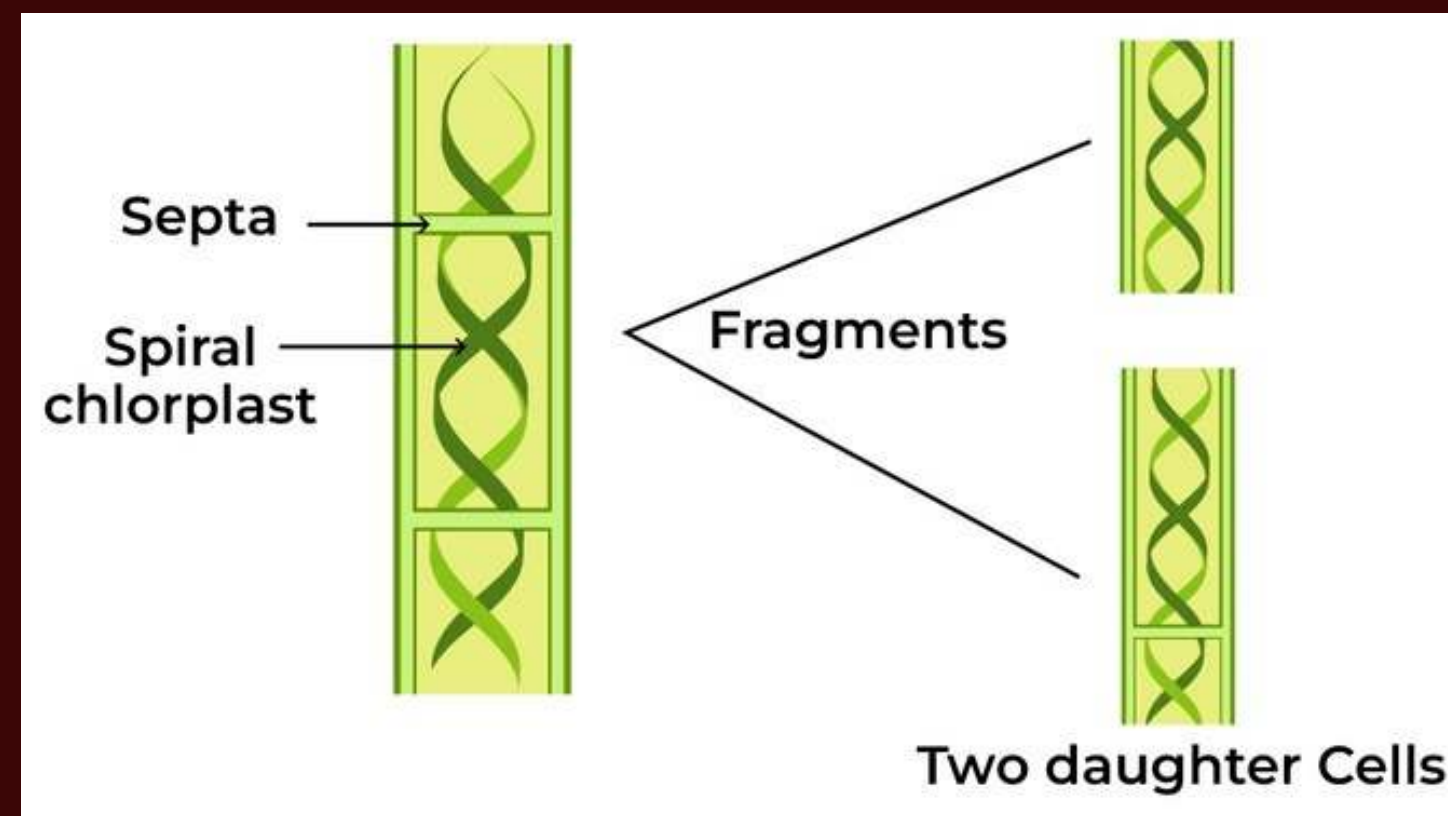
Fragmentation is a type of asexual reproduction in which an organism breaks up into smaller pieces upon maturation, and each piece develops into a new individual. This process occurs in simple multicellular organisms that can regenerate lost parts.

- Organisms like Spirogyra break into fragments, each growing into a new individual.
- Only works in simple organisms with less specialized cell organization.

How Fragmentation Occurs:

1. The mature parent organism breaks into fragments naturally or due to external factors like water currents.
2. Each fragment grows and develops into a complete organism.
3. This process leads to rapid population growth under favorable conditions.

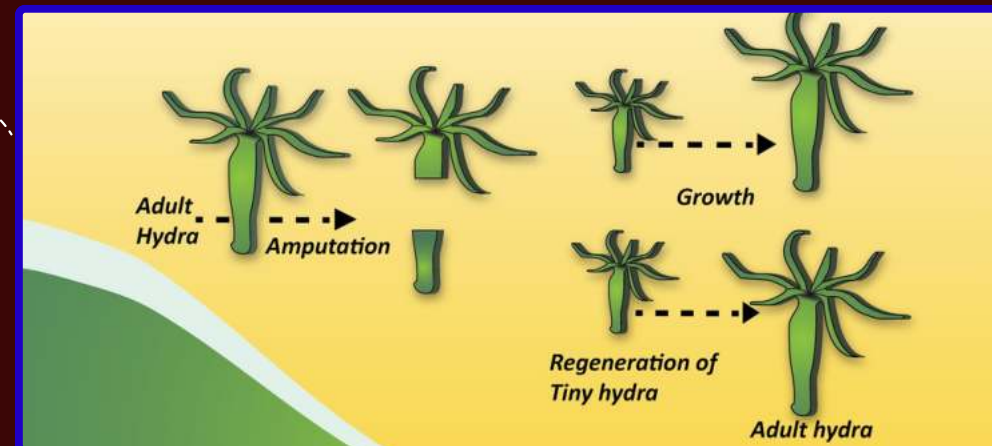
Spirogyra (a filamentous green algae) breaks into smaller pieces, and each fragment grows into a new individual.



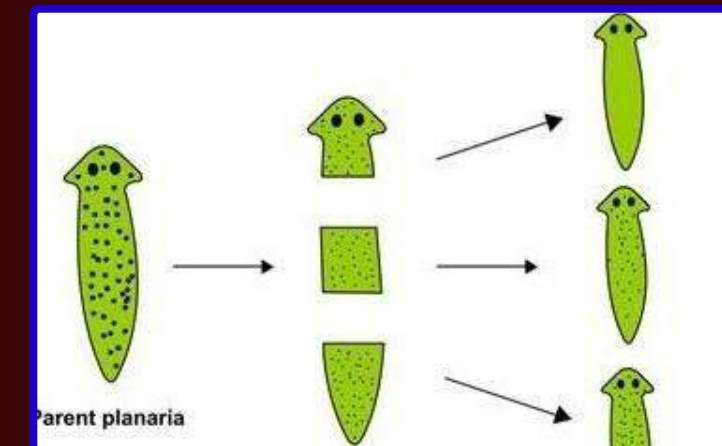
3. Regeneration

- Some organisms like Hydra and Planaria regrow entire bodies from cut parts.
- Performed by specialized cells that undergo developmental changes.
- Not a primary mode of reproduction, as it depends on external damage.

Regeneration in Hydra

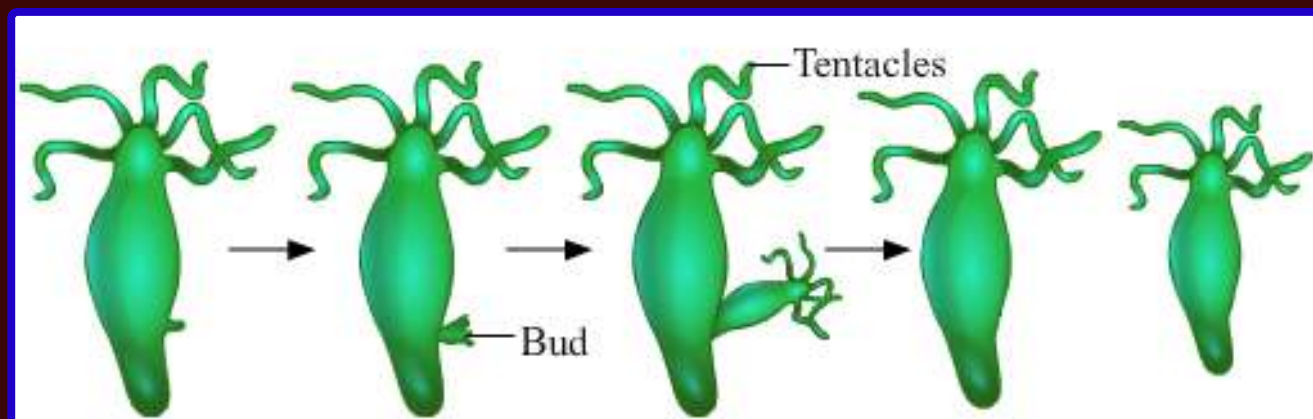


Regeneration in Planaria

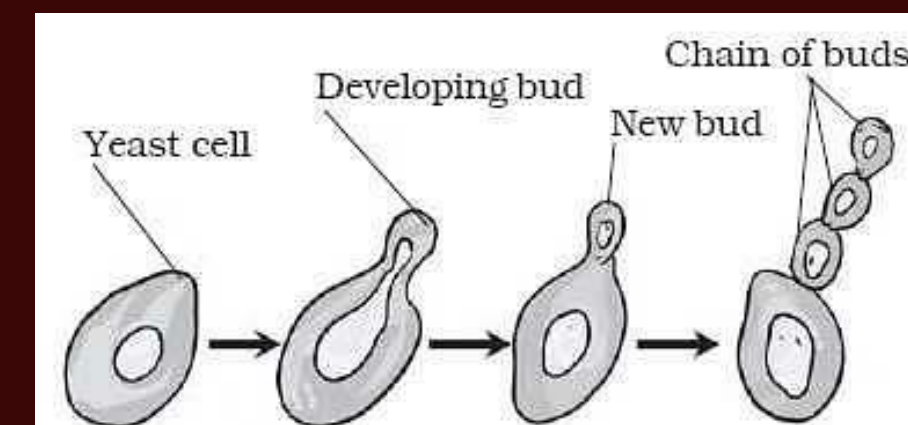


4. Budding

- Seen in organisms like Hydra.
- A new organism grows as a bud on the parent body, matures, and detaches to live independently.



Budding in hydra



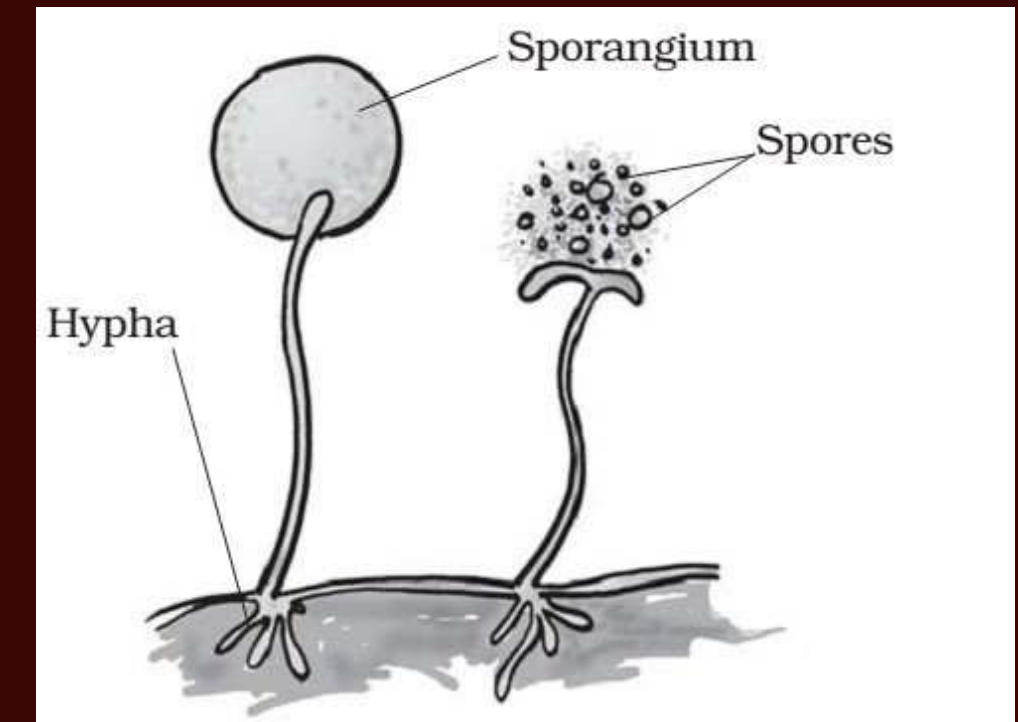
Budding in Yeast

5. Spore Formation

Spore formation is a type of asexual reproduction in which the parent organism produces tiny spores that grow into new individuals under favorable conditions.

How Spore Formation Occurs:

- The parent organism develops thread-like structures called hyphae (non-reproductive parts).
 - Tiny blob-on-a-stick structures appear, which are sporangia (spore-producing bodies).
 - Inside the sporangia, numerous spores are formed.
 - When mature, the sporangium bursts, releasing spores into the surroundings.
 - Under favorable conditions (moisture, warmth), each spore germinates into a new Rhizopus individual.
- Example: Rhizopus (bread mould) reproduces through spore formation. The hyphae grow on moist bread, while the sporangia release spores to form new mould colonies.



Spore formation in Rhizopus

Advantages of Spore Formation:

- **Survival in Harsh Conditions:** Spores are resistant to extreme heat, cold, and drought, ensuring survival.
- **Wide Dispersal:** Spores are lightweight and spread by wind, water, or animals, helping colonization.
- **Rapid Population Growth:** Large numbers of spores are produced, ensuring species survival despite losses.

6. Vegetative Propagation (Plants)

The method of producing new plants from vegetative parts like roots, stems, and leaves.

Reproductive part

Vegetative part

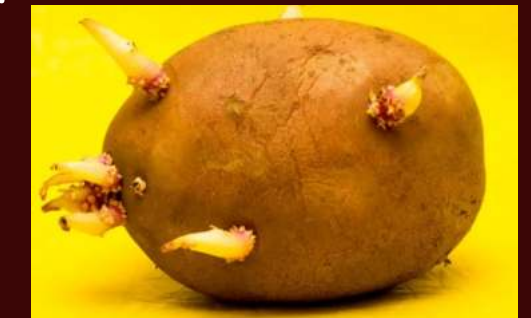
This part of the plant (flower) which is involved in the process of sexual reproduction.

These parts of the plant (stem, leaf, root) are not involved in the process of sexual reproduction.

Natural vegetation:

refers to plants that grow without human intervention. New plants develop from the leaves, roots, or stems of the parent plant.

- **By Underground Stem:** Tubers: Swollen, fleshy underground stems with buds (eyes). Example: Potato (Buds grow into new plants).
- **By Root:** Tuberous roots: Fleshy roots with buds that detach and grow into new plants. Examples: Sweet Potato, Dahlia, Tulip.
- **By Leaves:** Leaf buds: Adventitious buds grow on leaf margins, fall off, and develop into plantlets. Examples: Bryophyllum, Kalanchoe.

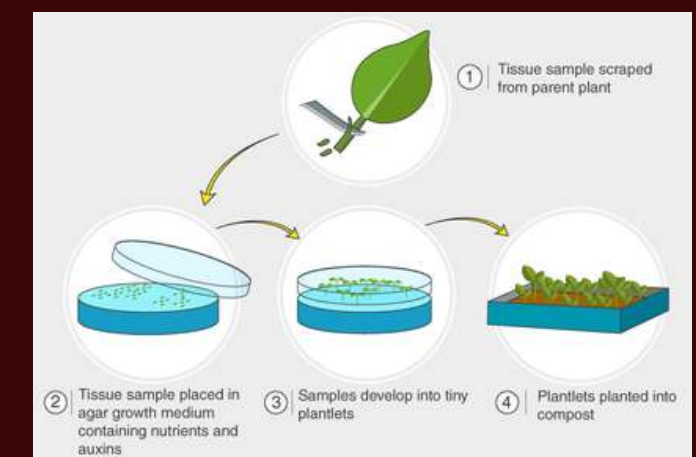
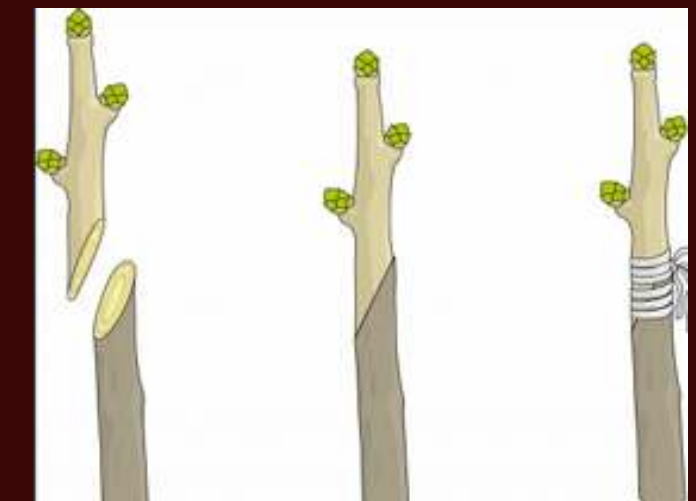


Artificial Vegetative Propagation :

Artificial vegetative propagation is a method of asexual reproduction in which new plants are grown artificially from the stems, roots, or leaves of a parent plant with human intervention.

Methods of Artificial Vegetative Propagation:

- ◆ **Cutting:** A cut portion of stem, root, or leaf is planted in soil, where it develops roots and grows into a new plant. Example: Rose, Hibiscus, Sugarcane.
- ◆ **Layering:** A lower branch of a plant is bent and covered with soil while still attached to the parent plant. After rooting, it is cut off and grows independently. Example: Jasmine, Strawberry.
- ◆ **Grafting:** A part of one plant (scion) is joined to the root system of another plant (stock) to combine desired traits. Example: Mango, Apple, Rose.
- ◆ **Tissue Culture:** A few plant cells are grown in a nutrient medium under controlled conditions to develop into a complete plant. Example: Orchid, Banana. Used for disease-free plant production (e.g., ornamental plants).



Advantages of Artificial Vegetative Propagation:

- **Faster Growth:** Plants grow faster than from seeds.
- **Genetic Similarity:** Offspring are identical to the parent plant.
- **Desirable Traits:** Maintains good quality plants (e.g., disease resistance, better yield).

Q.A feature of reproduction that is common to Amoeba, Yeast and Spirogyra is that

- (a) they reproduce asexually**
- (b) they are all unicellular**
- (c) they reproduce only sexually**
- (d) they are all multicellular**

Q. A Planaria worm is cut horizontally in the middle into two halves P and Q such that the part P contains the whole head of the worm. Another Planaria worm is cut vertically into two halves R and S in such a way that both the cut pieces R and S contain half head each. Which of the cut pieces of the two Planaria worms could regenerate to form the complete respective worms?

- (a) Only P**
- (b) Only R and S**
- (c) P, R and S**
- (d) P, Q, R and S**

Q. The ability of a cell to divide into several cells during reproduction in Plasmodium is called

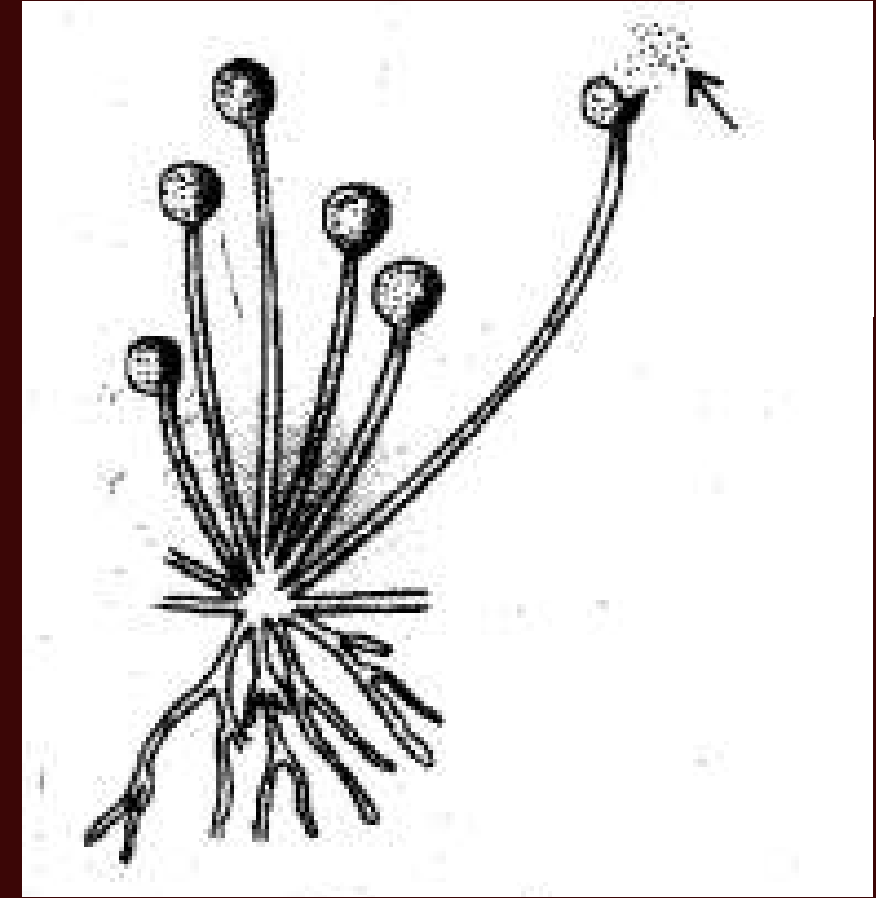
- (a) budding**
- (b) multiple fission**
- (c) binary fission**
- (d) reduction division**

Q. Identify the mode of asexual reproduction in the following organisms:



- (a) Fragmentation**
- (b) Multiple Fission**
- (c) Budding**
- (d) Binary Fission**

Q. Name and explain the process taking place in the given diagram.

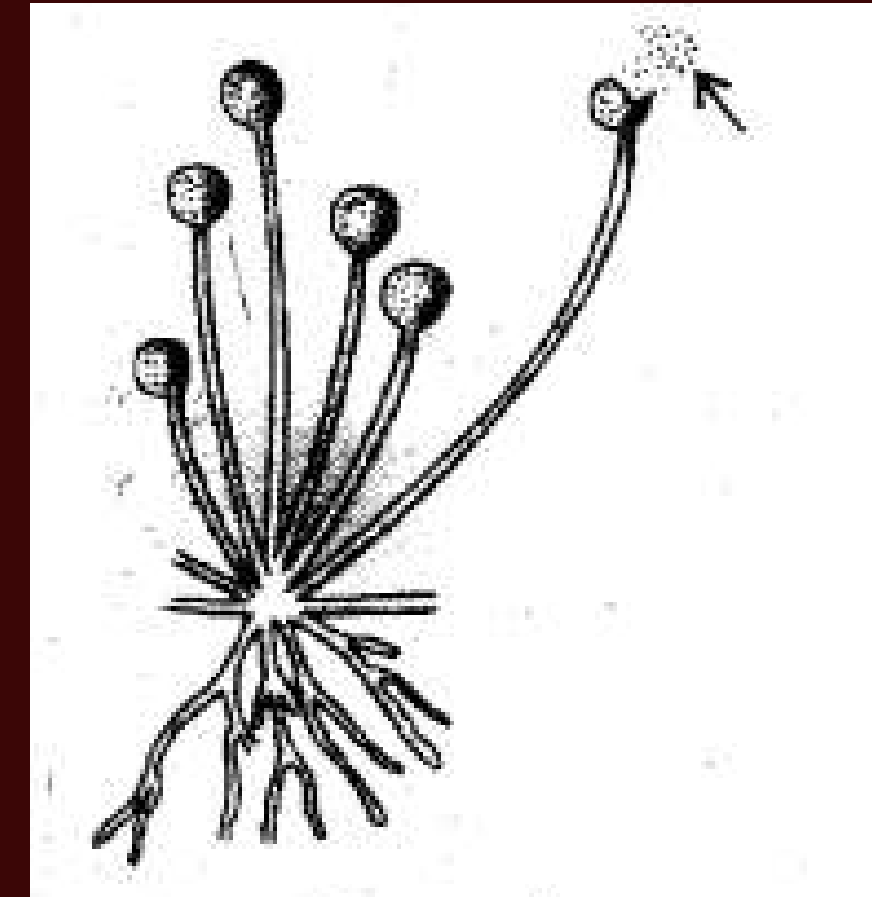


Q. Name and explain the process taking place in the given diagram.

Answer. The process shown in the diagram is **Spore Formation**, an asexual mode of reproduction.

Explanation:

- Rhizopus (bread mold) reproduces through spore formation.
- It produces spores inside sporangia (blob-like structures present at the tips of hyphae).
- When the sporangium matures, it bursts, releasing spores into the surroundings.
- These spores are lightweight and dispersed by wind, water, or animals.
- Under favorable conditions, spores germinate and develop into new individuals.



Q. What happens when

(a) accidentally, Planaria gets cut into many pieces-

(b) Bryophyllum leaf falls on the wet soil

(c) on maturation sporangia of Rhizopus bursts?

Q. What happens when

(a) accidentally, Planaria gets cut into many pieces-

(b) Bryophyllum leaf falls on the wet soil

(c) on maturation sporangia of Rhizopus bursts?

Answer: (a) When Planaria accidentally gets cut into many pieces then its each piece grows into a complete organism. This is known as regeneration. (b) When the Bryophyllum leaf falls on the wet soil, the buds present in the notches along the leaf margin develop into new plants. This is known as vegetative propagation. (c) The sporangia of Rhizopus contain cells or spores that can eventually develop into new Rhizopus individuals when it bursts on maturation.

Q. What is vegetative propagation? Write two of its advantages.

Answer: Vegetative propagation is a method of asexual reproduction in plants where new plants are produced from the roots, stems, leaves, or buds. Advantages include:

- It helps preserve the characters of the plants through successive generations as the genetic character is the same for the parent plant and offspring.**
- It allows for the propagation of plants that do not produce viable seeds**

Q. (a) Identify the modes of asexual reproduction in each of the following organisms:

(i) Hydra

(ii) Planaria

(iii) Amoeba

(iv) Spirogyra

(v) Rhizopus

(b) List three advantages of vegetative propagation.

(c) Why cannot fertilisation take place in flowers if pollination does not occur?

Q. (a) Identify the modes of asexual reproduction in each of the following organisms:

(i) Hydra (ii) Planaria (iii) Amoeba (iv) Spirogyra (v) Rhizopus

(b) List three advantages of vegetative propagation.

(c) Why cannot fertilisation take place in flowers if pollination does not occur?

Answer: (a) (i) Hydra – Budding

(ii) Planaria – Regeneration

(iii) Amoeba – Fission

(iv) Spirogyra – Fragmentation

(v) Rhizopus – Spores

(b) Advantages of vegetative propagation are : It is a quick method of propagation. The new plants produced by artificial vegetative propagation are exactly like the parent plants. Many plants can be grown from one plant by vegetative propagation. (c) Pollination brings male gametes in close proximity to the female reproductive part of flower. In the absence of pollination, there will be no male gametes available for fertilisation with female gametes.

Q. During a biology experiment, a student observed two different types of asexual reproduction in organisms. In the first case, a hydra was seen developing a small bud on its side, which eventually detached to form a new individual. In the second case, an amoeba was observed undergoing a process where it elongated and divided into two equal parts, resulting in two new amoebas.

Questions:

- 1. Identify the two types of asexual reproduction observed in the case study.**
- 2. Explain the process of budding in hydra and binary fission in amoeba, highlighting the steps involved in each process.**
- 3. Draw and label neat diagrams to illustrate both budding in hydra and binary fission in amoeba.**
- 4. Discuss the advantages of asexual reproduction in organisms.**

Answer.

1. Types of Asexual Reproduction: Budding (in hydra), Binary fission (in amoeba)

2. Process Explanation:

Budding in Hydra:

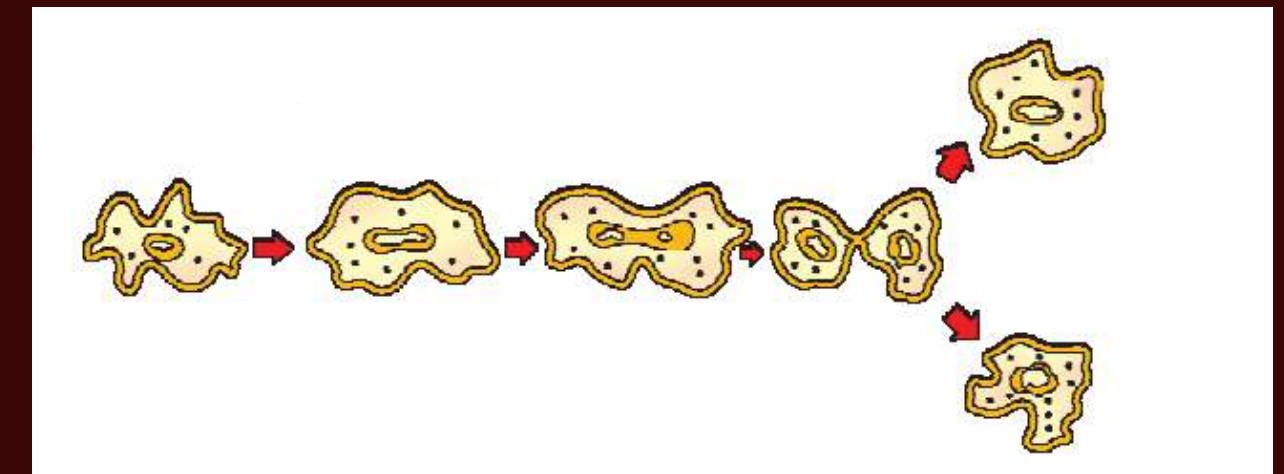
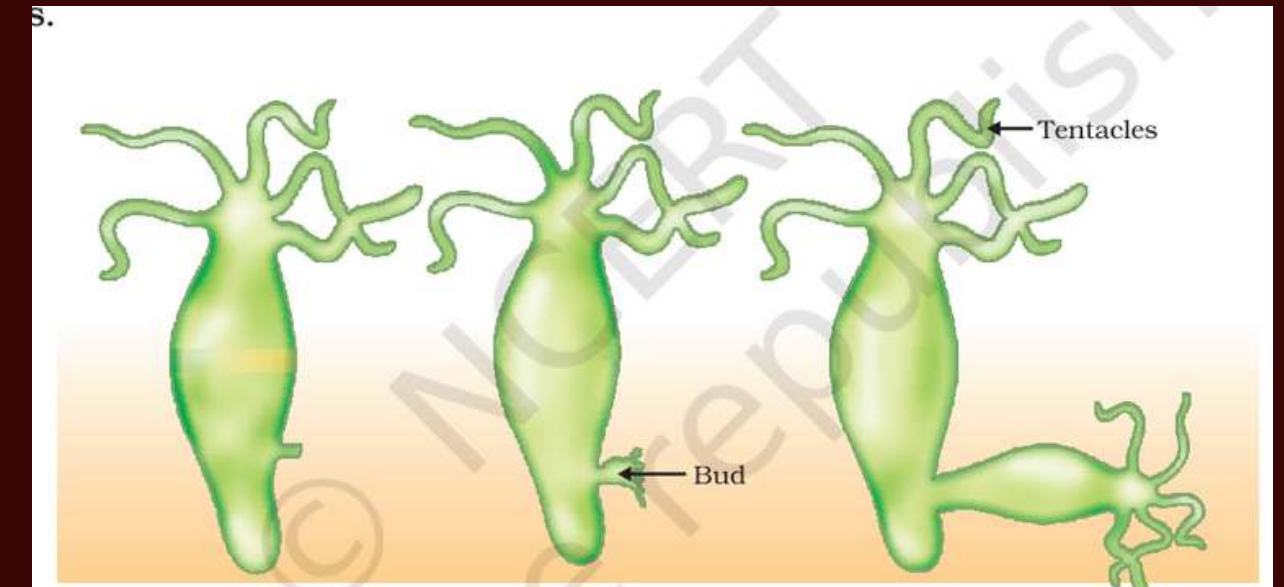
- A small bud forms on the side of the hydra.
- The bud grows and develops tentacles and a mouth.
- Eventually, the bud detaches, becoming a new individual.

Binary Fission in Amoeba:

- The amoeba's nucleus divides by mitosis.
- The cell elongates and the cytoplasm begins to split.
- The cell membrane pinches inwards, dividing the amoeba into two equal parts.

4. Advantages of Asexual Reproduction:

- Rapid population increase.
- Does not require a mate, making reproduction easier in isolated environments.
- Offspring are genetically identical, which can be beneficial in stable environments.



SEXUAL REPRODUCTION

It is a mode of reproduction that depends on the involvement of two individuals before a new generation can be created.

Steps in sexual reproduction

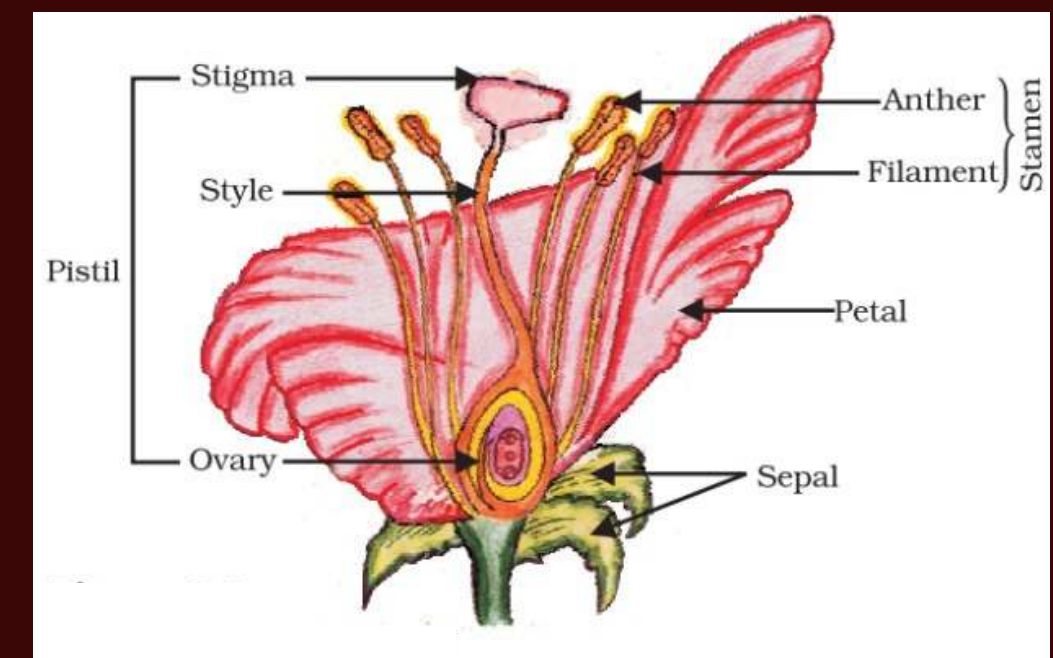
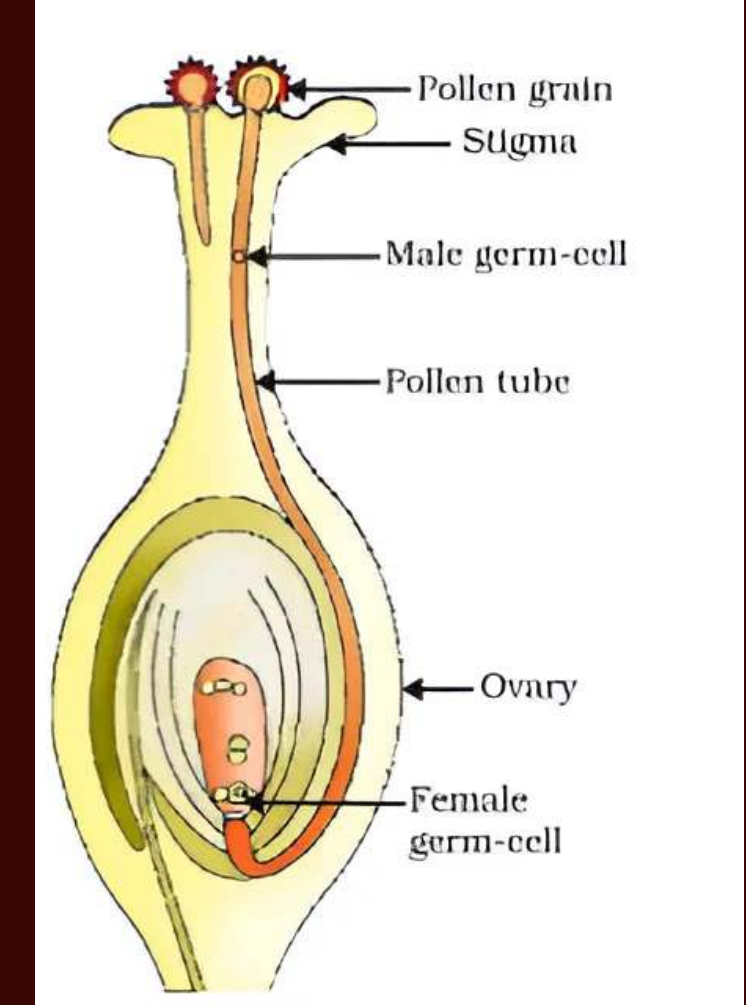
- Formation of gametes in the sex organs.
- Transfer of male gamete to female gamete.
- Fusion of gametes, either inside or outside the female parent's body.
- Development of the zygote to embryo and then complete individual.

Advantages of sexual reproduction

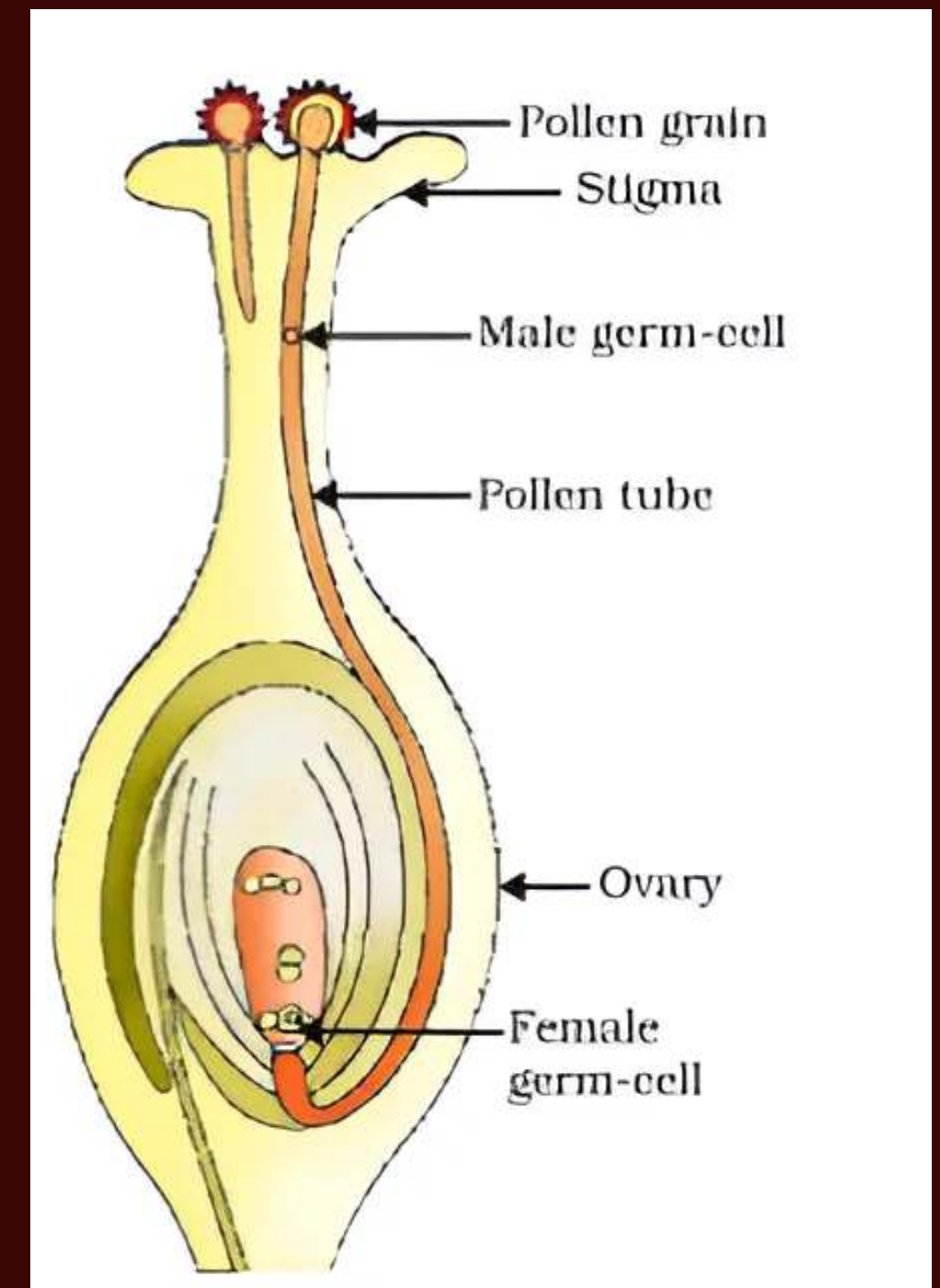
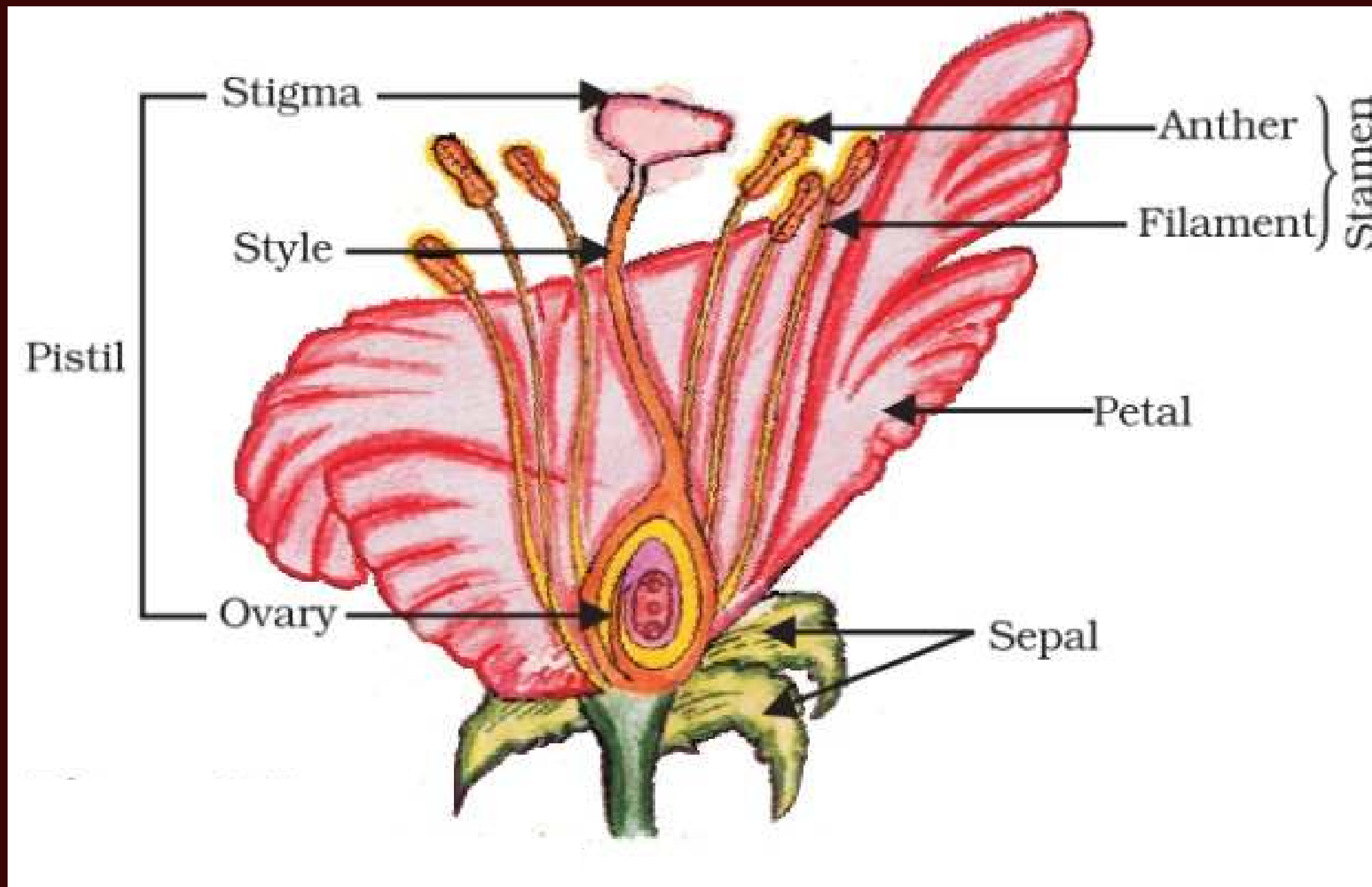
- Variations are produced among the progeny.
- Such populations are able to adapt well to changing environment and thus evolves faster.

Sexual Reproduction in Flowering Plants

Parts	Function
SEPALS	Usually green and provide protection to flower during bud stage
PETALS	Brightly coloured and have strong fragrance to attract pollinators
ANTHER	Produces pollen grain which consists male gametes
FILAMENT	It forms the stalk that bears anther
STIGMA	Receives pollen grain during pollination
STYLE	Elongated structure, connects stigma and ovary, pollen tube travel through the style to reach the ovule.
OVARY	Basal swollen part of pistil, converts into fruit after fertilization.
OVULE	Present inside ovary, consists of female gamete, site of fertilization.



Sexual Reproduction in Flowering Plants



Reproduction in plants

Stamen/Male reproductive organ

Pistil/Carpel/Female reproductive organ

BISEXUAL: Both stamen and pistil are present in a same flower. E.g., Rose, lily

UNISEXUAL: either stamen or pistil is present. E.g., Papaya, Watermelon,

Pollination: Transfer of pollen grains from the anther of a stamen to the stigma of a pistil. This can happen through wind, insects, or other agents.

AGENTS OF POLLINATION

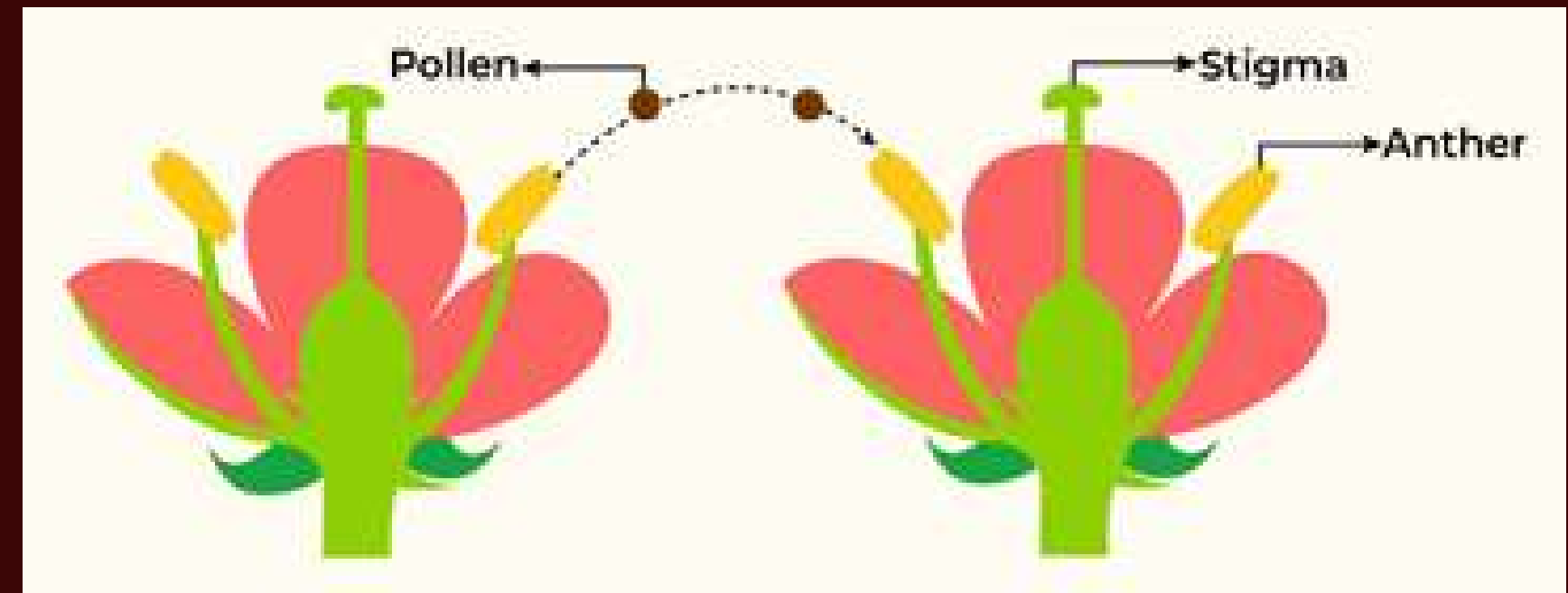
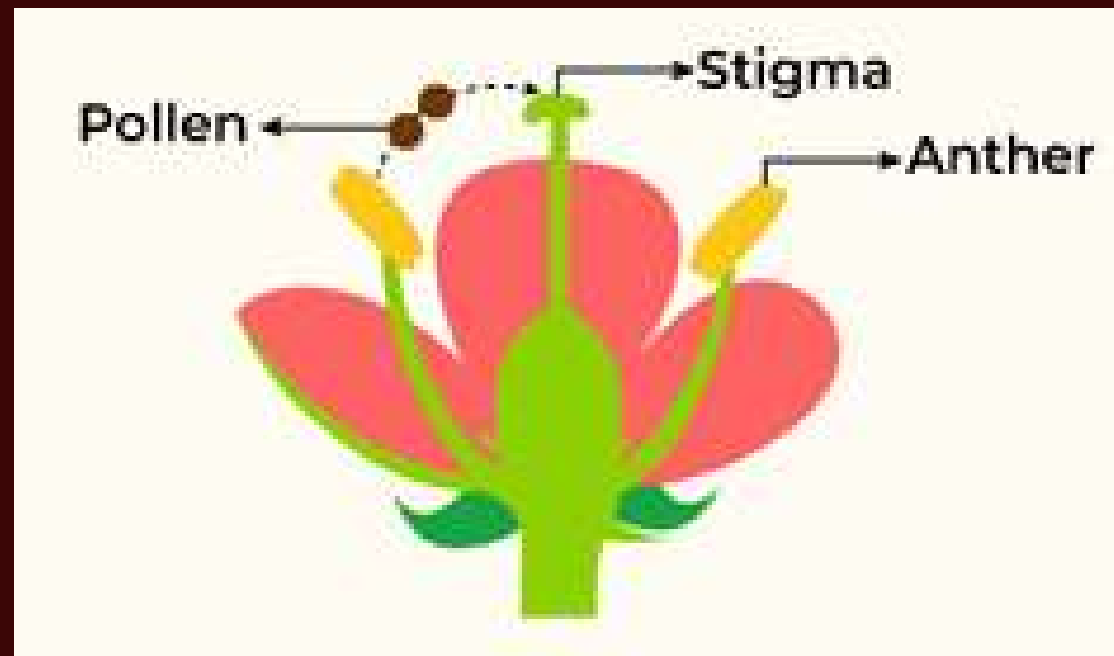
Abiotic

Example - Wind and water

Biotic

Example - Birds, Insects, Bats, and snails

- **Self-pollination** – Pollen transfers within the same flower.
- **Cross-pollination** – Pollen transfers between different flowers via wind, water, or animals.

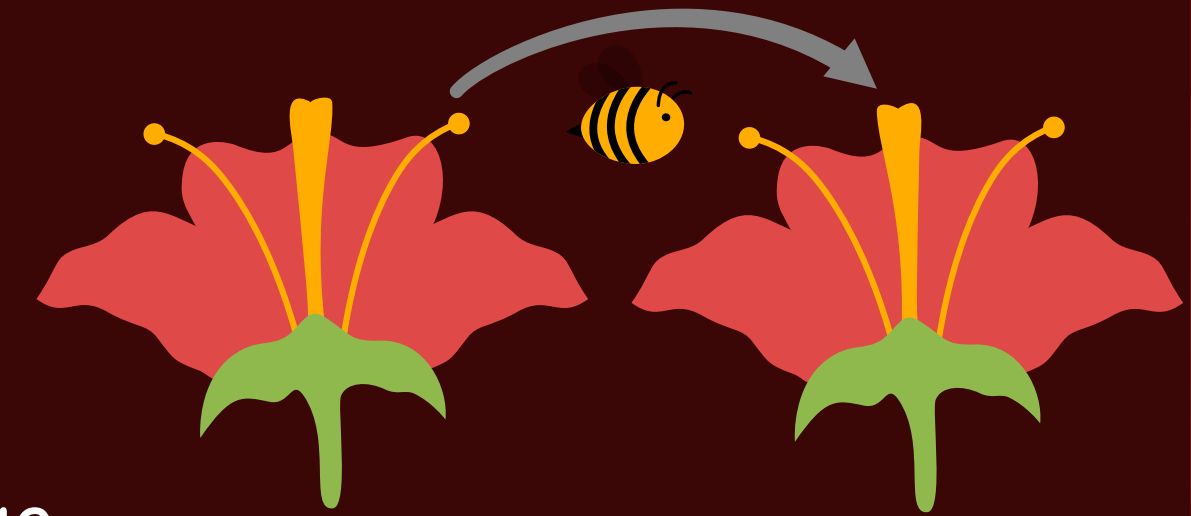


Fertilization – Pollen germinates, a pollen tube forms, and male germ-cells fuse with female gametes in the ovary to form a zygote. Pollination: Pollen grain carrying the male gamete lands on the stigma.

- **Pollen Tube Formation:** The pollen grain germinates, forming a pollen tube that grows towards the ovule.
- **Male Gamete Transfer:** The male gamete travels through the pollen tube and reaches the ovule via the micropyle.
- **Fertilization:** The male gamete fuses with the female gamete inside the ovule, forming a zygote.

Post-fertilization Changes –

- Petals, sepals, stamens, style, and stigma wither and fall.
- The ovule matures into a seed.
- The ovary develops into a fruit.
- The endosperm provides nutrients for the developing embryo.



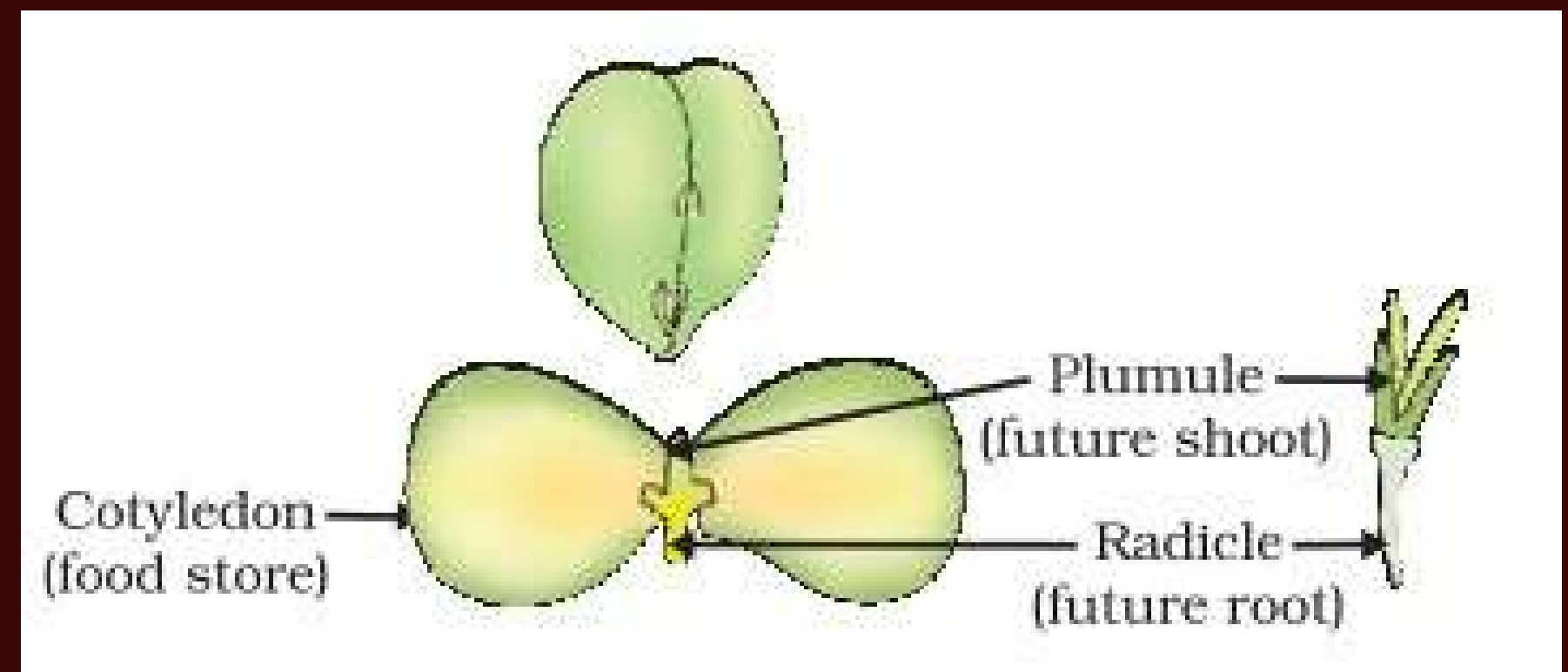
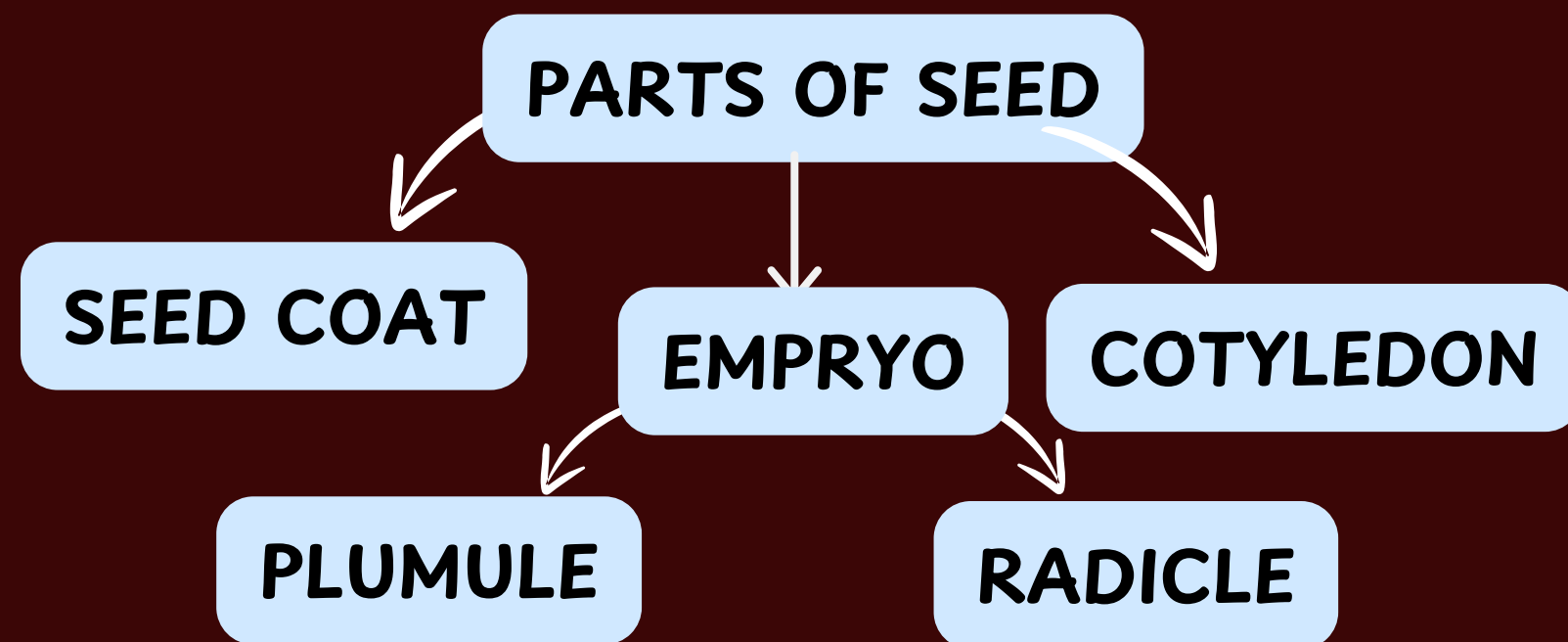
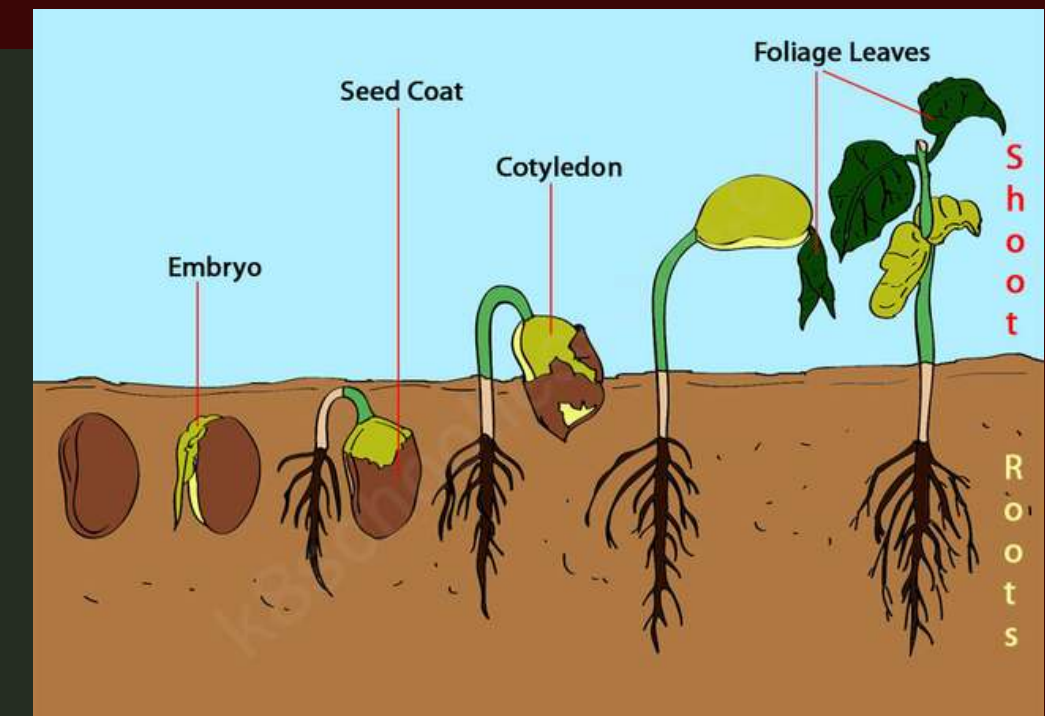
Pollination	Fertilization
Transfer of pollen grain from anther to stigma of a flower.	Fusion of male gamete with female gamete.
Achieved by agents like wind, water, or animals.	Achieved by the growth of the pollen tube, allowing the male gamete to reach the female germ cell.
Leads to fertilization.	Leads to the formation of seeds.
External process.	Internal process.

Seed is the final product of Sexual reproduction.

Seed develops into embryo which then develops into seedling under favorable conditions.

Parts of Seed:

- **Seed coat** - It is the protective layer.
- **Cotyledon** - Food storage & provides nourishment to growing embryo.
- **Embryo**
 - **Plumule** - Shoot system
 - **Radicle** - Root system



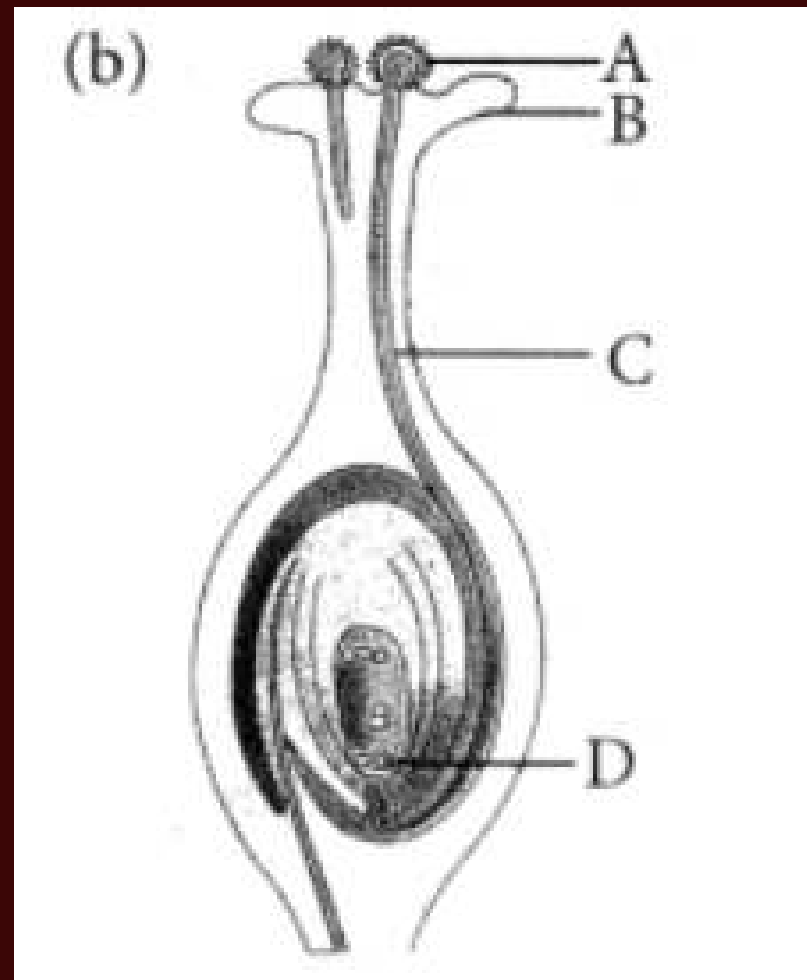
Q. Part(s) of a flower which attracts insects for pollination is (are)

- (a) Petals and Sepals**
- (b) Anther and Stigma**
- (c) Petals only**
- (d) Sepals only**

Q. Length of the pollen grain depends upon the distance between -----.

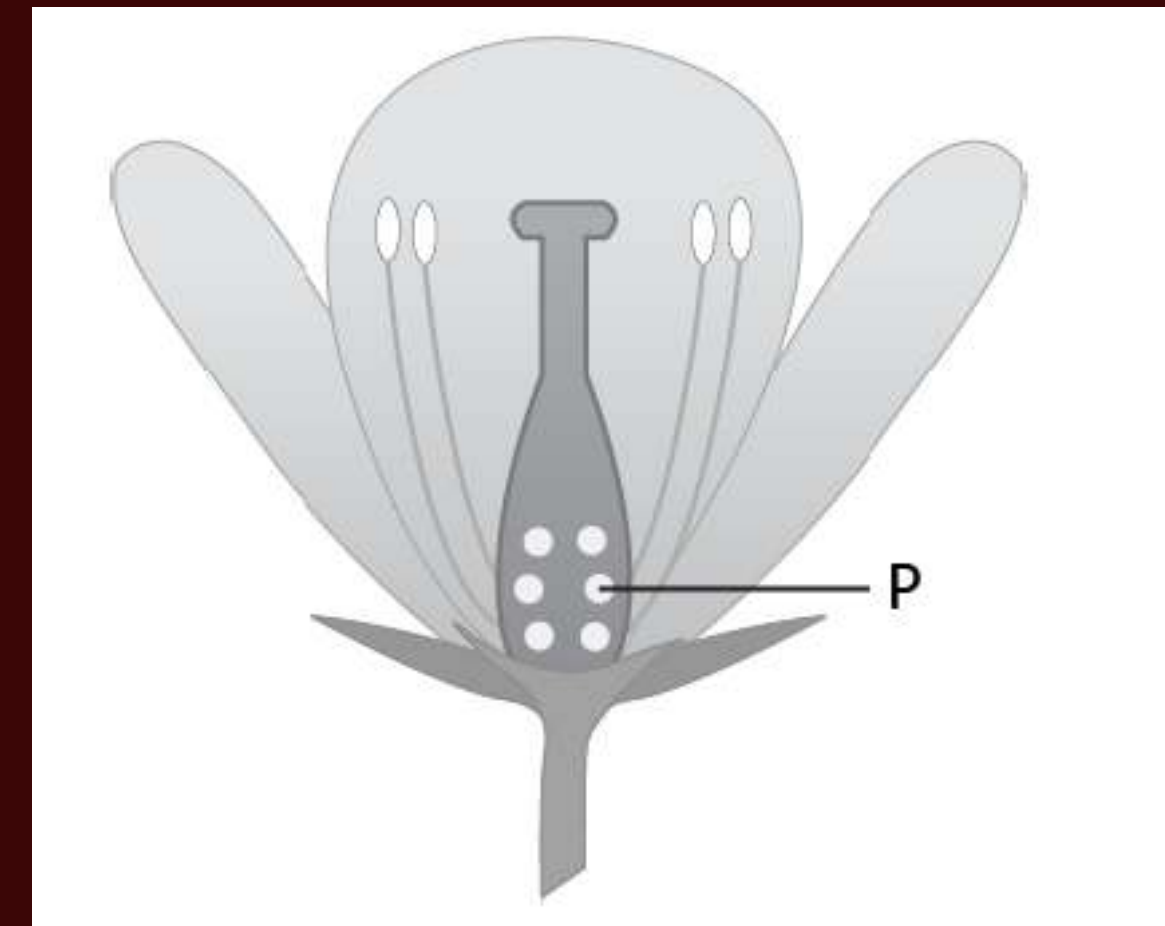
- (a) pollen grain and upper surface of stigma**
- (b) pollen grain on upper surface of stigma and ovule**
- (c) pollen grain in anther and upper surface of stigma**
- (d) upper surface of stigma and lower part of style**

- Q. (i) Name the part marked A in the diagram.
- (ii) How does A reaches part B?
- (iii) State the importance of the part C.
- (iv) What happens to the part marked D after fertilisation is over?



Q. The image shows the structure of a flower. Which of the following processes will likely be disturbed or not occur if the labelled part is removed from the flower?

- a) Formation of fruit**
- b) Transport of pollen**
- c) Formation of pollen**
- d) Development of the pollen tube**



Q. Differentiate between self-pollination and cross-pollination. Also state agents of Pollination and Their Role.

Q. Differentiate between self-pollination and cross-pollination. Also state agents of Pollination and Their Role.

Answer. Self-Pollination: Transfer of pollen from the anther to the stigma of the same flower or another flower of the same plant.

- **Cross-Pollination:** Transfer of pollen from the anther of one flower to the stigma of another flower on a different plant of the same species.

Agents of Pollination and Their Role:

- **Wind** – Carries lightweight pollen grains from one flower to another (e.g., Maize).
- **Insects** – Collect nectar and transfer pollen while moving from flower to flower (e.g., Sunflower).

Q. Explain the process of fertilization in flowering plants. What happens to the ovary and ovule after fertilization?

Q. Explain the process of fertilization in flowering plants. What happens to the ovary and ovule after fertilization?

Answer. After pollination, the pollen grain lands on the stigma and germinates, forming a pollen tube that extends through the style and reaches the ovary. The male gamete travels through the pollen tube and fuses with the female gamete (egg cell) inside the ovule, resulting in the formation of a zygote. This process is called fertilization.

What Happens After Fertilization?

- The ovule develops into a seed. 🌱
- The ovary matures into a fruit, which protects the seed. 🍏

Reproduction in Human Beings

Puberty is the period during which sexual maturation occurs, leading to reproductive capability.

Changes During Puberty:

Common in both genders: Growth of thick hair in armpits and genital areas, oily skin, pimples, and increased awareness of bodily changes.

- **Girls:** Breast development, menstruation starts, darkening of nipple tips.
- **Boys:** Growth of facial hair, deepening of voice, enlargement of penis, occasional erections.

Why does puberty occur?

The body prioritizes growth first; reproductive tissues mature later as body growth slows down.

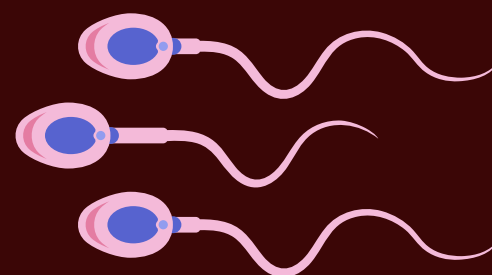
Sexual Maturity Indicators: Body changes (hair growth, body structure) signal readiness for reproduction.

Role of Male & Female Organs:

- Penis is specialized for sperm transfer.
- Female reproductive system matures for pregnancy and breastfeeding.

Males

- Sperm cell
- Male sex hormone



Females

- Egg cell
- Female Sex hormones

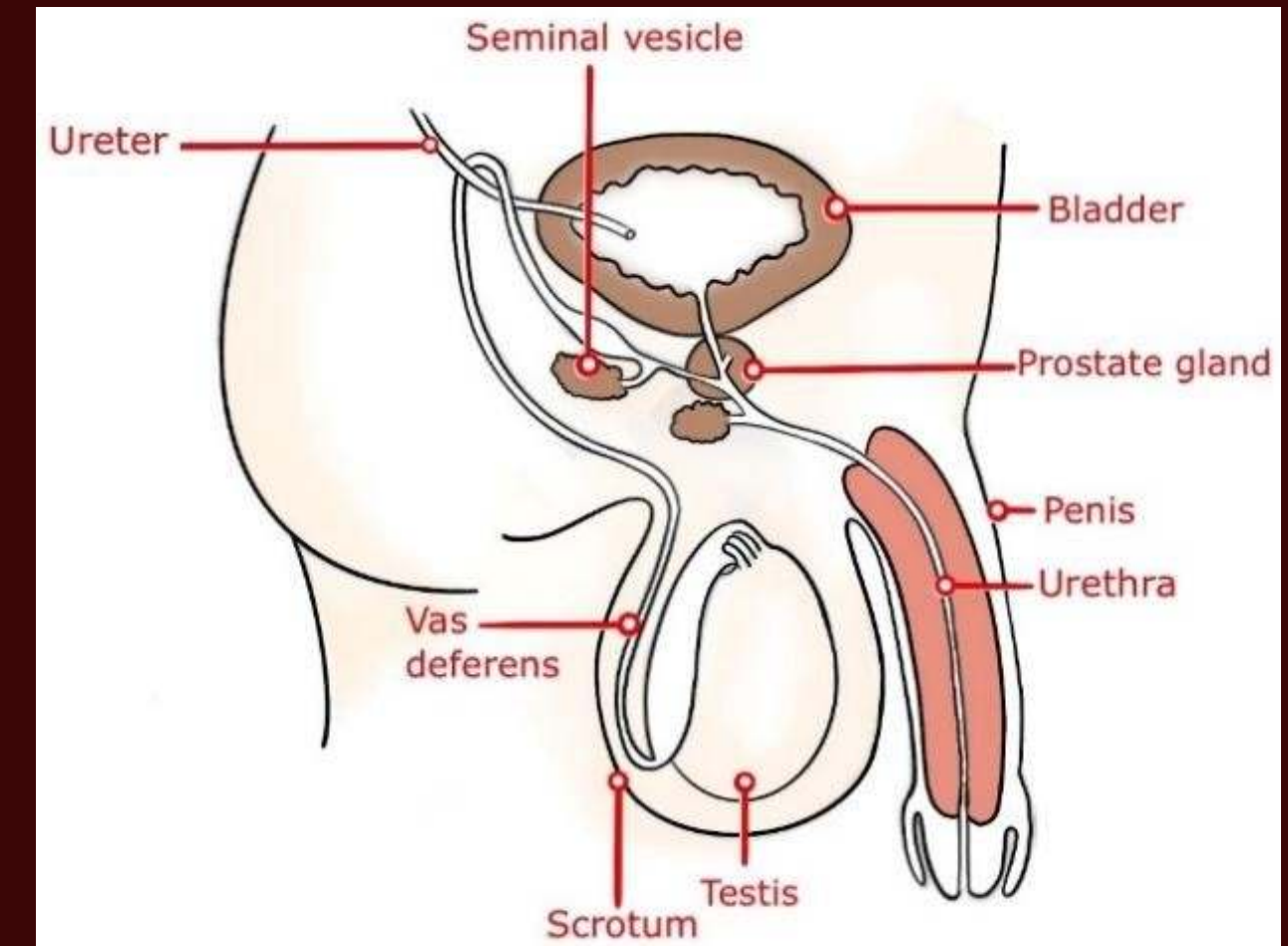


Male Reproductive System

Organ	Location/Structure	Function
Testes	Located in the scrotum, outside the abdominal cavity.	Produce sperm and secrete testosterone.
Scrotum	Sac-like muscular structure outside the abdominal cavity.	Maintains a lower temperature for sperm production.
Vas deferens	Tube connecting the epididymis to the urethra.	Transports sperm to the urethra during ejaculation.
Seminal vesicle	Located near the base of the bladder.	Secretes fluid that nourishes sperm and forms part of semen.
Prostate gland	Below the bladder, around the urethra.	Secretes fluid that protects sperm from acidity.
Bladder	Located in the pelvic region.	Stores urine.
Urethra	Tube extending from the bladder to the penis.	Conveys sperm and urine for discharge.
Penis	External organ of the male reproductive system.	Deposits sperm in the vagina and disposes of urine.

Functions of the Male Reproductive System:

- Produces germ cells (sperms).
- Delivers sperms to the site of fertilization.



🔥 Key Concepts:

- **Scrotum helps maintain a lower temperature essential for sperm production.**
- **Testosterone controls male puberty and sperm production.**
- **Sperms are carried in semen, which aids transport and nutrition.**
- **Urethra serves as a common passage for both urine and sperms.**

Q. Which among the following is not the function of the testes at puberty?

(i) Formation of germ cells

(ii) Secretion of testosterone

(iii) Development of placenta

(iv) Secretion of estrogen

a) (i) and (ii)

b) (ii) and (iii)

c) (iii) and (iv)

d) (i) and (iv)

Q. (a) What is puberty?

(b) Describe in brief the functions of the following parts in the human male reproductive system.

(i) Testes (ii) Seminal vesicle (iii) Vas deferens (iv) Urethra

1. Why are testes located outside the abdominal cavity?

2. State how sperms move towards the female germ cell.

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1. Why are testes located outside the abdominal cavity?

2. State how sperms move towards the female germ cell. (2020)

Ans: (a) The age at which the sex hormones begin to be produced and the boy and girl becomes sexually mature, i.e., able to reproduce is called puberty.

(b) (i) Testes: The two testes in male are the sites where male gametes, i.e., sperms are formed. Testes also produce the male sex hormone called testosterone.

(ii) Seminal vesicles: Seminal vesicles are one pair of sac-like structures near the base of bladder. Seminal fluid is a watery alkaline fluid that contains nutrients (fructose) which serve as a source of energy for the sperm. Each seminal vesicle releases its contents into the ejaculatory duct during ejaculation. (iii) Vas deferens: This is a straight tube, about 40 cm long, which carries the sperms to the seminal vesicles, where mucus and a watery alkaline fluid containing fructose, mix with the sperms. (iv) Urethra : It is a long tube that arises from urinary bladder. Urethra carries urine from the bladder as well as sperms from the vas deferens, through the penis.

(c) Testes are located outside the abdominal cavity because sperm formation requires a lower temperature than normal body temperature. The temperature of the testes in the scrotum is about 2-

2.5°C lower than normal body temperature. This temperature is ideal for sperm formation and development.

(d) The sperms present in the testes of man are introduced into the vagina of the woman through penis during copulation. Millions of sperms are released into the vagina at one time. The sperms are highly active and mobile. They travel from here upward through the uterus at the top of fallopian tube within five minutes.

Q. List two roles of each of the following in human reproductive system:

(i) Seminal vesicles and prostate gland

(ii) Testis

Q. List two roles of each of the following in human reproductive system:

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(ii) Testis

Answer. (i) Seminal vesicles and prostate gland:

- 1. Seminal vesicles secrete a fluid that provides nutrients to sperms.**
- 2. The prostate gland secretes a fluid that helps in the transport of sperms.**

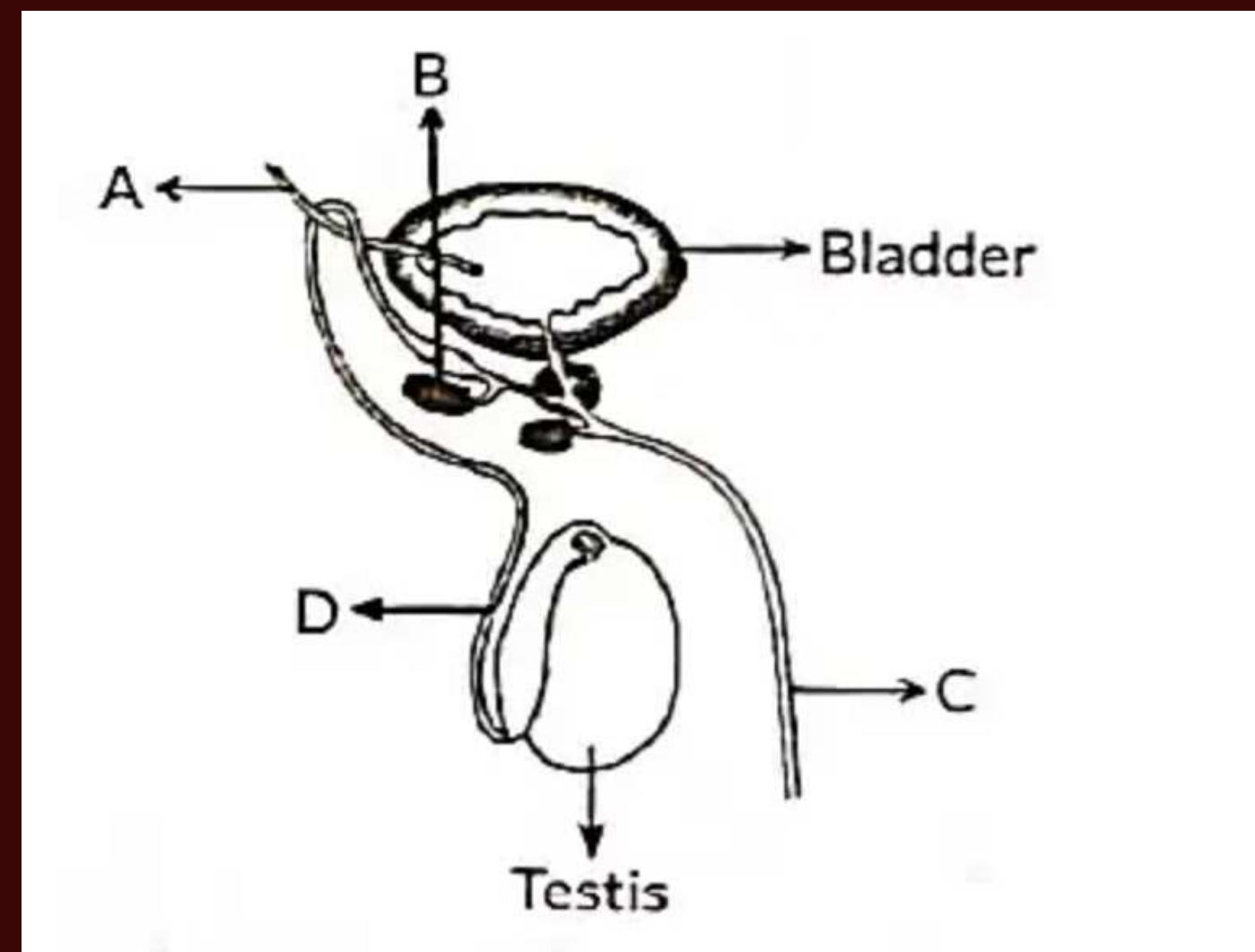
(ii) Testis:

- 1. It produces sperms, the male gametes.**
- 2. It secretes the male hormone testosterone, which regulates secondary sexual characteristics.**

Q. Based on the given diagram answer the questions given below:

(a) Label the parts A, B, C and D.

(b) Name the hormone secreted by testis and mention its role. (c) State the functions of B and C in the process of reproduction.



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(a) Label the parts A, B, C and D.

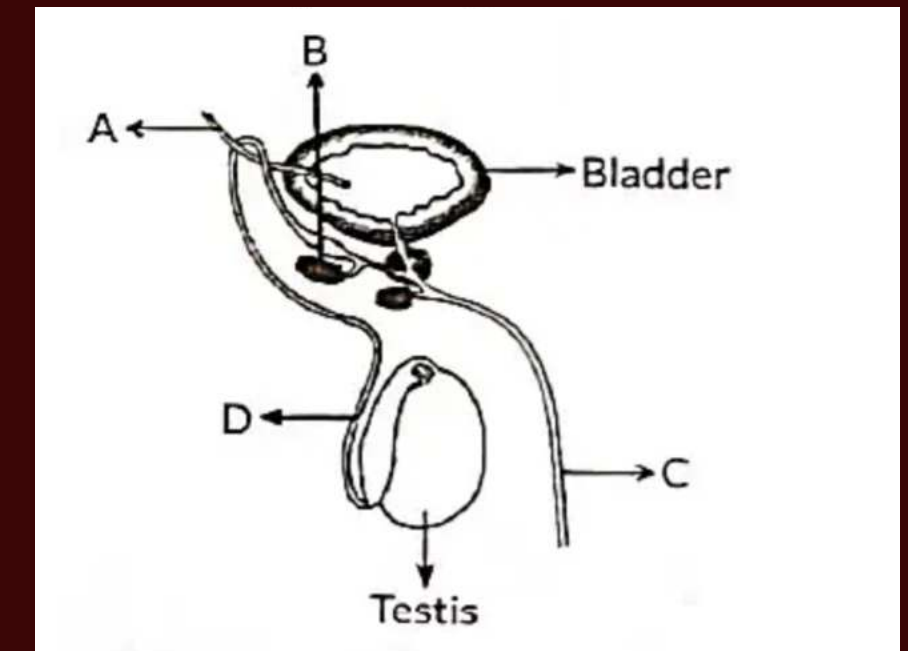
(b) Name the hormone secreted by testis and mention its role. (c) State the functions of B and C in the process of reproduction.

Ans: (a) A-Ureter

B - Seminal vesicle

C - Urethra

D - Vas deferens



(b) Testes produce male sex hormone testosterone. Hormone testosterone brings about the development of secondary sexual characters during puberty in boys like growth of facial hair, deepening of voice, build up of muscle mass and also regulates formation of sperms.

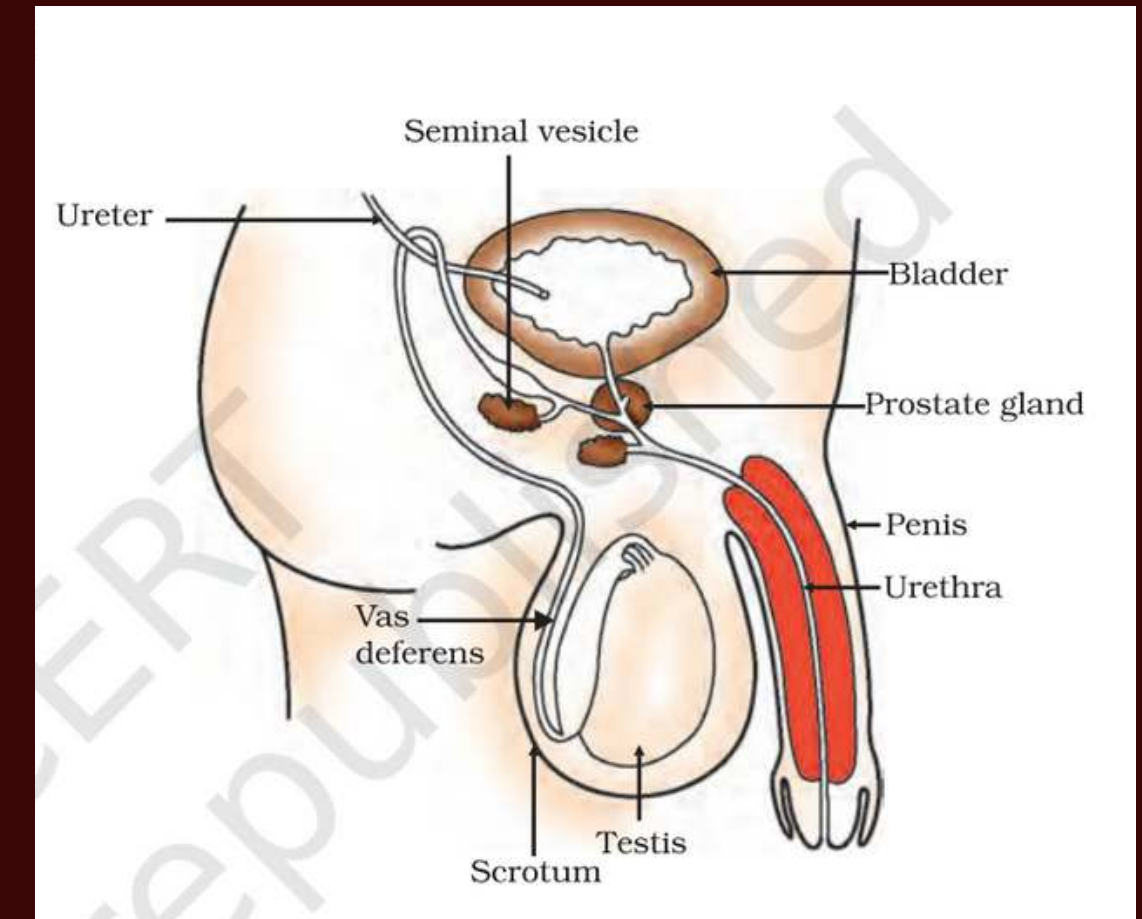
(c) Seminal vesicles (B) release its contents into the ejaculatory duct during ejaculation. Urethra (C) carries sperms from the vas deferens through the penis.

Q. Draw a neat and labeled diagram of the human male reproductive system. Write its main functions.

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Answer. Main Functions of the Human Male Reproductive System:

- **Sperm Production:** The testes produce sperm cells (germ cells) in the seminiferous tubules.
- **Hormone Secretion:** The testes secrete testosterone, which regulates sperm formation and causes secondary sexual characteristics during puberty.
- **Sperm Transport:** The vas deferens transports sperm from the testes to the urethra.
- **Seminal Fluid Production:** Glands such as the prostate and seminal vesicles add fluids to sperm, providing nourishment and facilitating transport.
- **Common Passage:** The urethra serves as a passage for both urine and sperm.

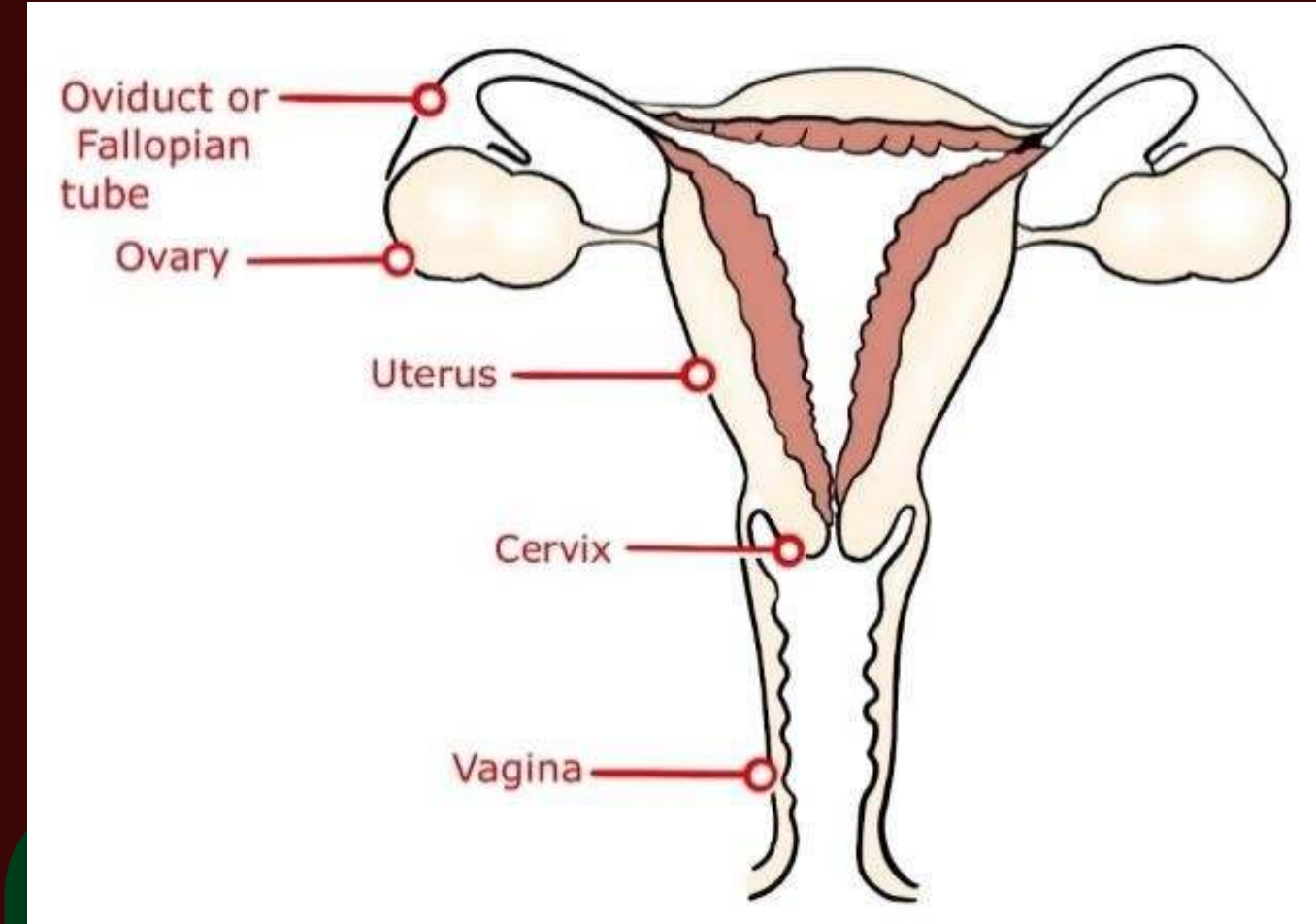


Female Reproductive System

Functions of the Female Reproductive System:

- Produces female germ cells (eggs or ova).
- Provides a site for fertilization and supports embryo development.
- Secretes female reproductive hormones.

Organ	Location/Structure	Function
Ovary	Paired organs located in the lower abdomen, on either side of the uterus.	Produces female gametes (ova/eggs) and secretes estrogen and progesterone.
Ovulation	The process in which an ovary releases an egg.	Release of an ovum from the ovary, typically occurring once a month.
Fallopian Tube	Tubes extending from the sides of the uterus.	Transports the egg to the uterus and provides the site for fertilization.
Uterus	Hollow, muscular organ located in the pelvic cavity.	Houses and supports the developing embryo; implantation occurs in the endometrium.
Cervix	Narrow, lower part of the uterus.	Connects the uterus to the vagina, acts as a passage for sperm, and dilates during childbirth.
Vagina	Muscular canal extending from the cervix to the external body.	Receives sperm during intercourse and serves as the birth canal during delivery.



🔥 Key Concepts:

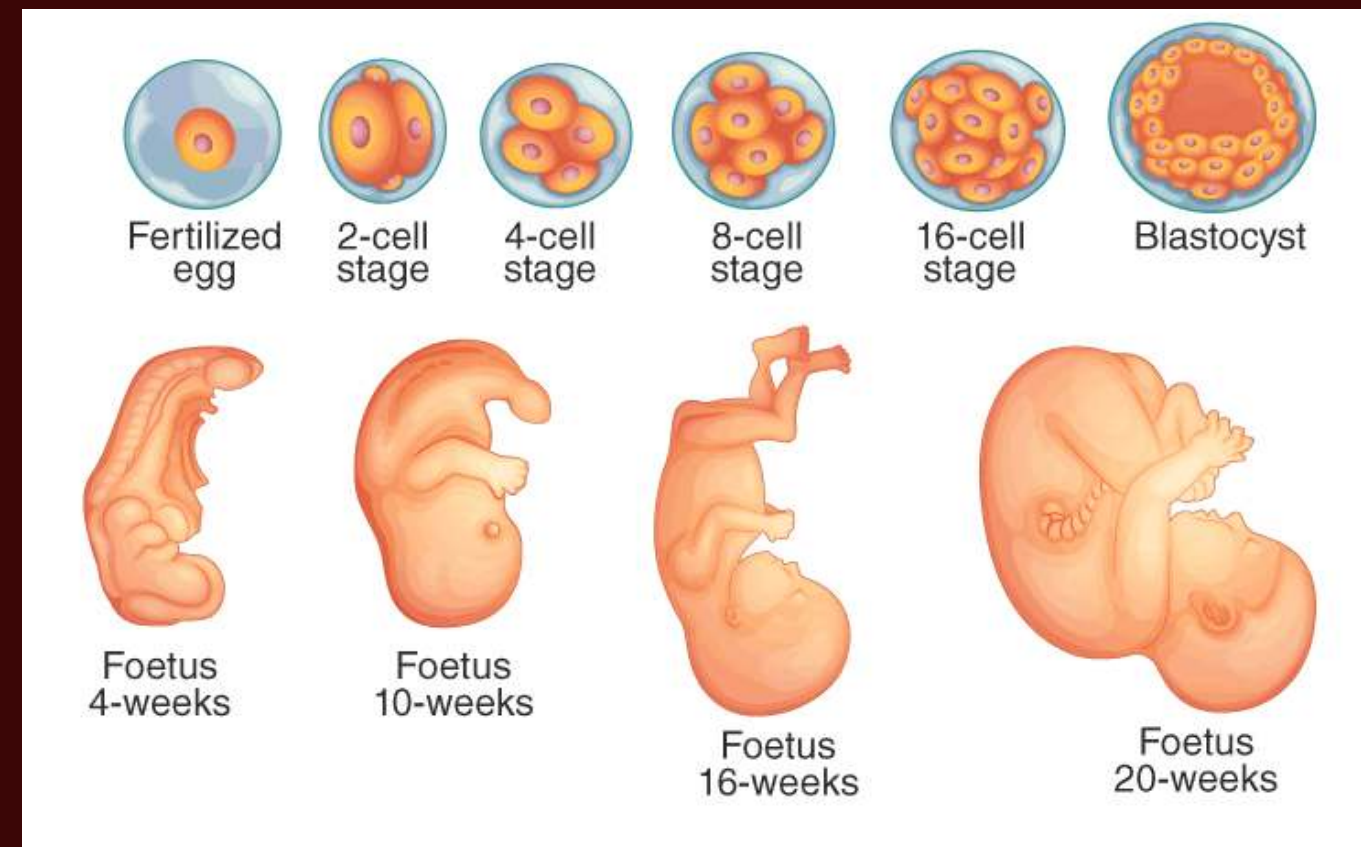
- **Ovaries produce eggs and hormones.**
- **Fertilization occurs in the fallopian tubes.**
- **The uterus supports embryo development.**
- **The placenta facilitates nutrient and waste exchange.**
- **Childbirth happens through uterine contractions.**
- **Menstruation occurs if the egg is not fertilized.**

Fertilization and Development of Embryo:

- Sperms enter the vagina during intercourse and travel to the fallopian tube.
- If sperm meets the egg, fertilization occurs, forming a zygote.
- The zygote divides to form an embryo, which implants in the uterus.
- The placenta (a special tissue) provides:
 - Nutrition (glucose, oxygen) from mother's blood.
 - Removes waste from the embryo.
- Pregnancy lasts about 9 months.
- Childbirth occurs through rhythmic contractions of the uterus.

Implantation: is the process where the fertilized egg (zygote) attaches to the lining of the uterus to develop into an embryo.

Implantation:



Nutrition to embryo:

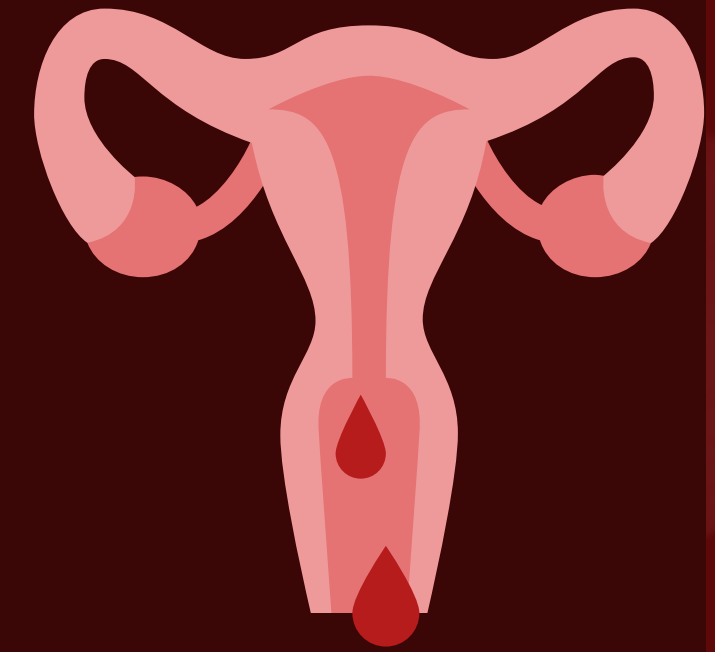
- A placenta is a special tissue connection between the embryo and the uterine wall.
- The lining of the uterus thickens and is richly supplied with blood to nourish the growing embryo.
- The embryo receives nutrients and oxygen from the mother’s blood through the placenta.
- It is embedded in the uterine wall. It contains villi on the embryo’s side of the tissue.
- This provides a large surface area for nutrition and oxygen to pass from the mother to the embryo.
- Waste substances produced by the embryo are removed through the placenta into the mother’s blood.



Zygote	Foetus
It is fertilized egg formed after the fusion of the sperm with the egg.	It is a state of the embryo that shows all the main recognizable body parts of a mature organism.
The zygote divides several times to form an embryo.	An embryo gradually develops into a foetus.

What Happens When the Egg is Not Fertilized?

- The egg survives for about one day if fertilization does not occur.
- The uterus prepares its lining every month for possible pregnancy.
- If the egg is not fertilized, the uterine lining breaks down and is expelled through the vagina.
- This process is called **menstruation**, which lasts for about 2 to 8 days.
- The menstrual cycle repeats roughly every 28 days.



MENSTRUATION:

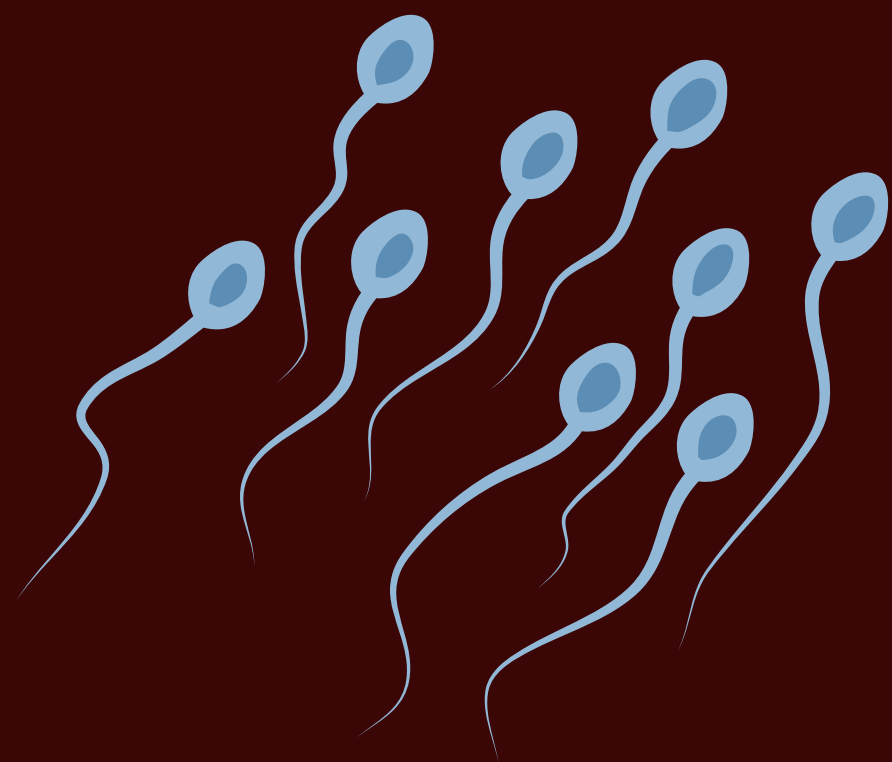
- The ovary releases one egg each month, causing the uterus to thicken its lining in preparation for potential fertilization.
- If the egg is not fertilized, it survives for about a day, and the uterine lining is no longer needed.
- This lining breaks down and is expelled through the vagina as blood and mucus.
- This monthly process, known as menstruation, typically lasts 2 to 8 days.

Menarche: 11 - 12 years

Menstruation: 12 - 50 years

Menopause: 50+ years

ASPECT	MALE GAMETE (SPERM)	FEMALE GAMETE (OVUM)
Size	Smaller	Larger than sperm
Motility	Motile (Swims using tail)	Non-motile
Production Site	Produced in testes	Produced in ovary
Quantity	Produced in large numbers	Only one is released per month



Q. (a) In the female reproductive system of human beings, state the functions of:

(i) Ovary (ii) Oviduct.

(b) Mention the changes which the uterus undergoes, when

(i) it has to receive a zygote. (ii) no fertilisation takes place.

(c) State the functions of placenta.



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(c) State the functions of placenta.

Ans: (a) (i) The ovaries in female are primary sex organs (or female gonads) which perform the dual function - production of female gametes (eggs or ova) and secretion of female sex hormones (estrogen and progesterone).

(ii) Oviducts or fallopian tube are paired tubes originating near the ovaries of their respective sides and extend upto uterus. The terminal part of fallopian tube is funnel-shaped with finger-like projections called fimbriae lying near ovary. Fimbriae pick up the ovum released from ovary and push it into fallopian tube.

Fertilisation also takes place in the oviduct.

(b) (i) As the ovary releases one egg every month, the uterus also prepares itself, every month to receive fertilised egg by making its lining thick and spongy to nourish the zygote if fertilisation takes place. (ii) When the female gamete/egg is not fertilised, this lining is not needed any longer. So, the lining slowly breaks and comes out through vagina as blood and mucus through menstrual cycle that takes place every month.

(c) Placenta performs the following functions:

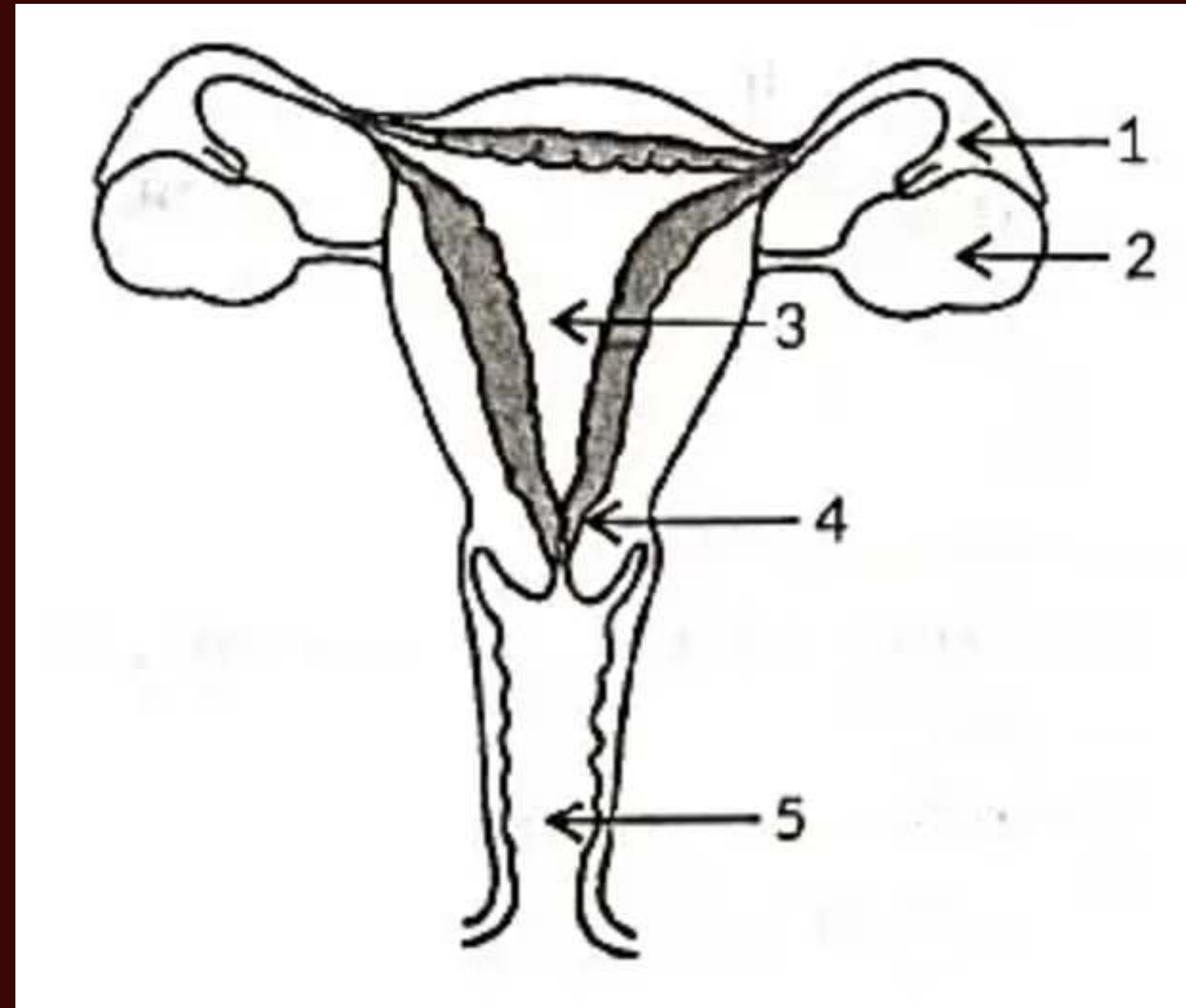
(i) All nutritive elements from maternal blood pass into the fetus through it.

(ii) Placenta helps in respiration i.e., supply of oxygen and removal of CO₂ from fetus to maternal blood. (iii)

Fetal excretory products diffuse out into maternal blood through placenta and are excreted by mother.

(iv) Placenta also secretes various hormones during pregnancy.

Q. Identify the given diagram. Name the parts 1 to 5



Q. Identify the given diagram. Name the parts 1 to 5

Ans: (a) The given diagram is the sectional view of human female reproductive system. The labelled parts are:

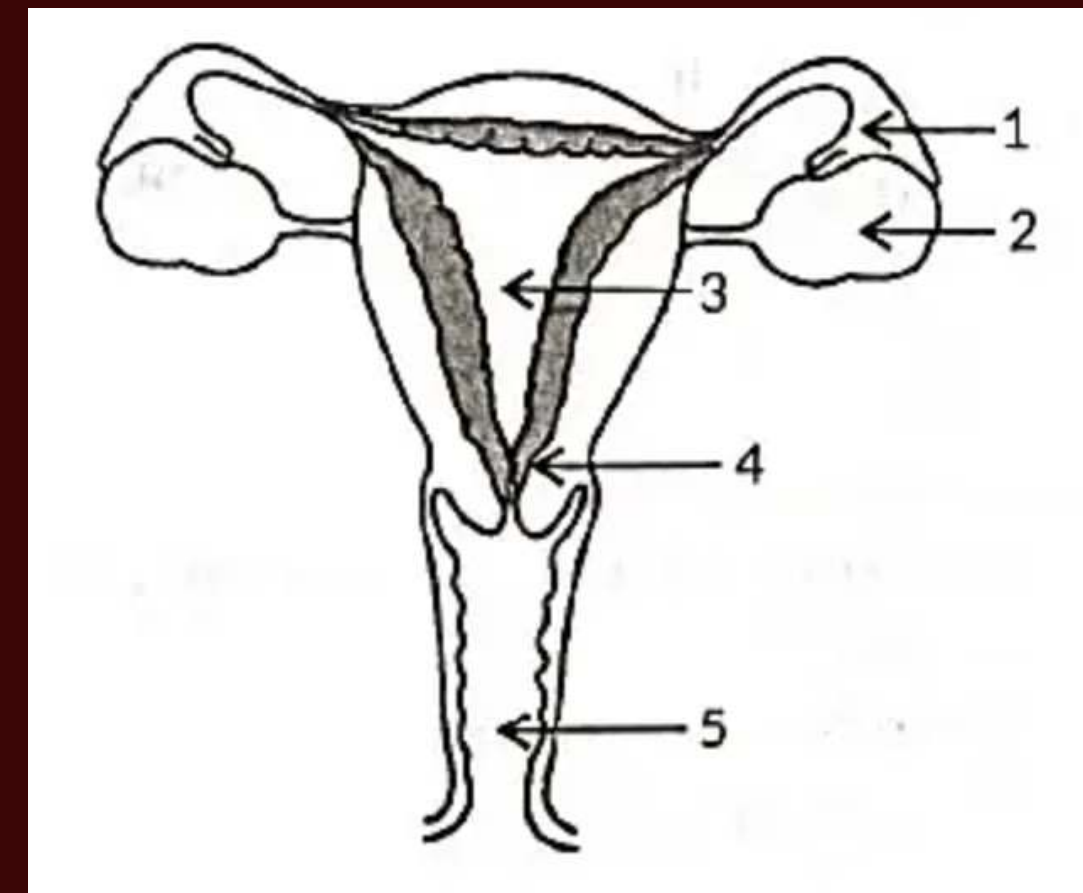
1. Funnel of fallopian tube or oviduct

2. Ovary

3. Uterus or womb

4. Cervix

5. Vagina



Q. (a) List two preparations shown every month by the uterus in anticipation of pregnancy in humans.

(b) Why is variation important in organisms?

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(b) Why is variation important in organisms?

Answer: (a) The two preparations shown every month by the uterus in anticipation of pregnancy in human are:

(i) the wall of uterus becomes thick to receive the fertilised egg and

(ii) the uterine wall is richly supplied with blood to nourish the growing embryo.

(b) Variation is important in organisms because:

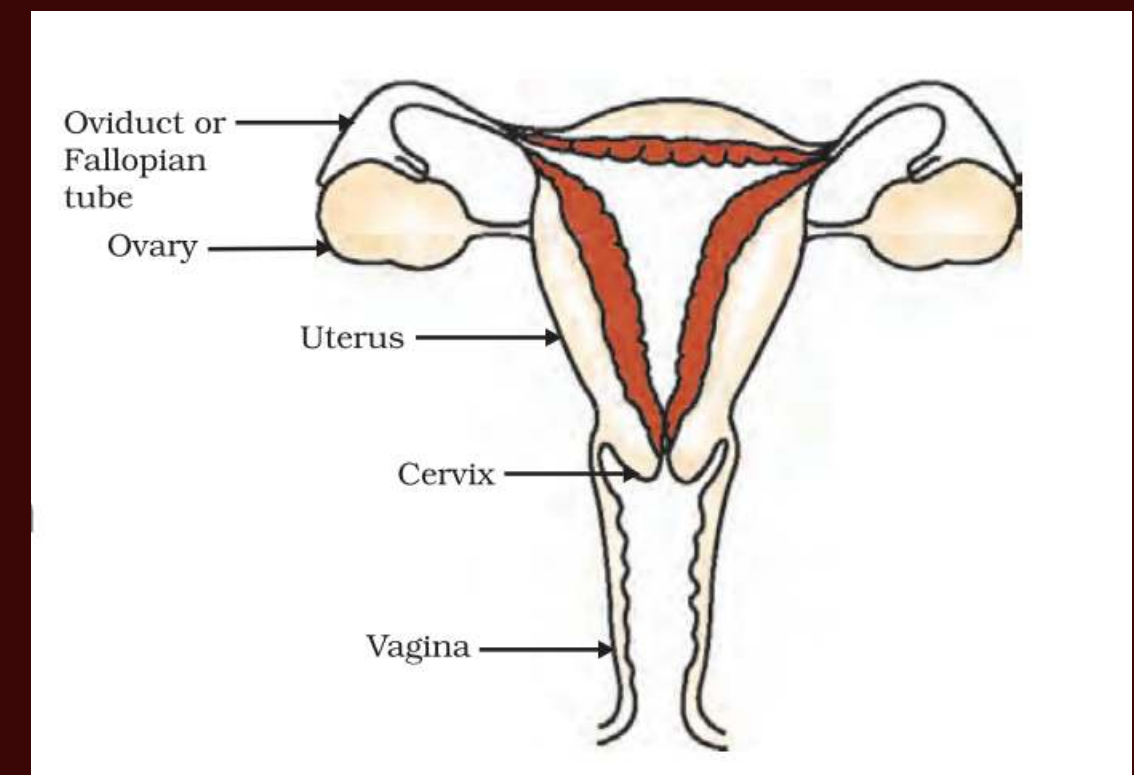
- Adaptation to Environment: Helps organisms survive environmental changes.**
- Foundation of Evolution: Drives natural selection and evolution of species.**
- Prevents Extinction: Increases chances of survival during diseases or climate changes.**
- Diversity in Traits: Leads to better chances of survival for the population.**

Q. Draw a neat and labeled diagram of the human female reproductive system. Write its main functions.

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Main Functions of the Human Female Reproductive System

1. **Egg Production:** The ovaries produce eggs (female germ cells) and secrete hormones like estrogen and progesterone.
2. **Egg Transport:** The oviducts (fallopian tubes) carry the matured egg from the ovaries to the uterus.
3. **Fertilization Site:** The oviducts are the site where sperm can meet and fertilize the egg.
4. **Embryo Development:** The uterus provides a suitable environment for the fertilized egg (zygote) to implant and develop into a fetus.
5. **Nutrient Transfer:** The placenta, formed from embryonic and maternal tissues, facilitates nutrient and waste exchange between the mother and the developing embryo.

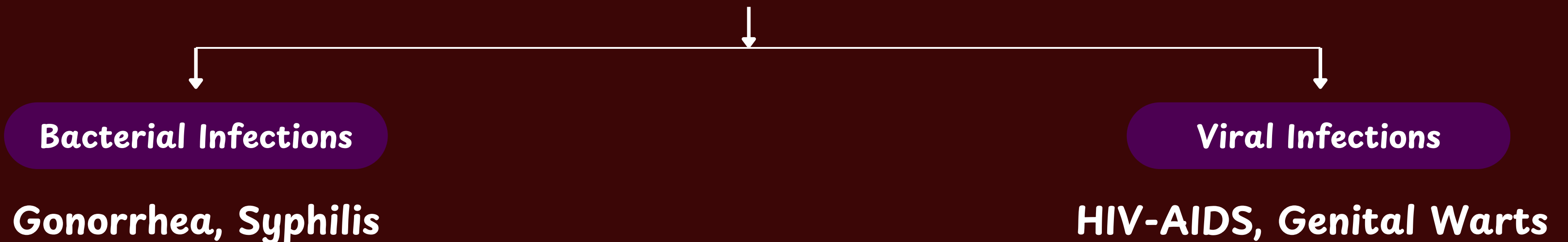


REPRODUCTIVE HEALTH

- It is the overall well-being of an individual in all aspects related to reproduction, including physical, emotional, and social health.
- It involves safe reproductive practices, awareness, and the prevention of diseases.

Sexually Transmitted Diseases (STDs):

Sexually transmitted diseases (STDs) are infections passed from one person to another through sexual contact with an infected individual.

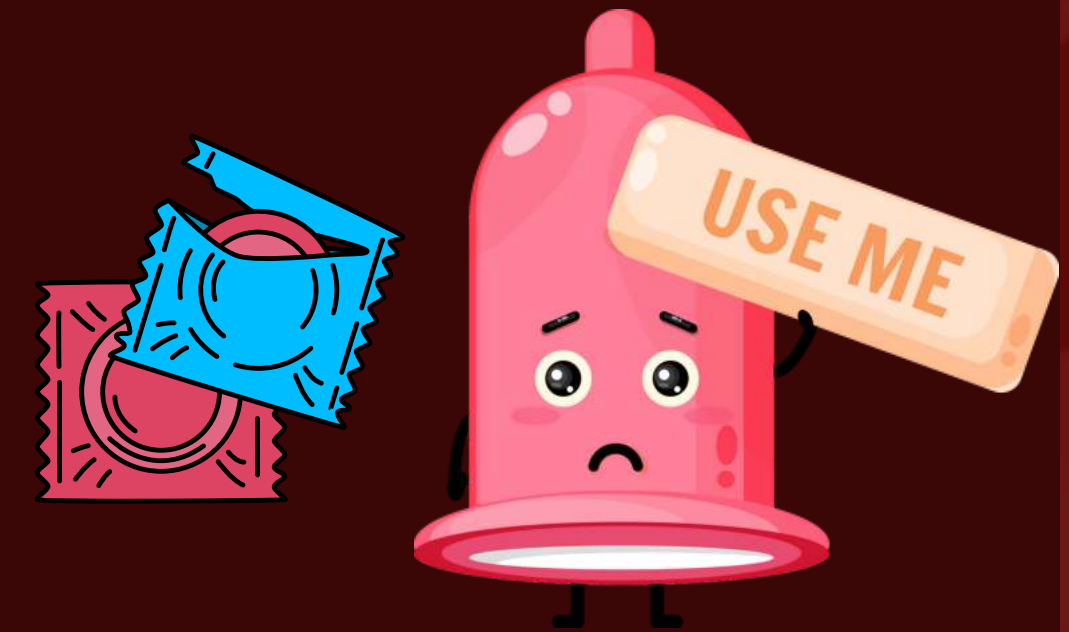


Contraception: Methods or devices used to prevent pregnancy. It includes techniques like condoms, oral pills, intrauterine devices (IUDs), and sterilization procedures. It should be adopted

- To prevent unwanted children.
- To maintain the distance between two children and plan the family
- To prevent sexually transmitted diseases.

1. Mechanical/barrier method

- Barrier methods, such as condoms (for males and females) and diaphragms (for females), prevent the sperm and ovum from meeting.
- They also help prevent the transmission of sexually transmitted diseases by blocking the intermixing of body fluids.



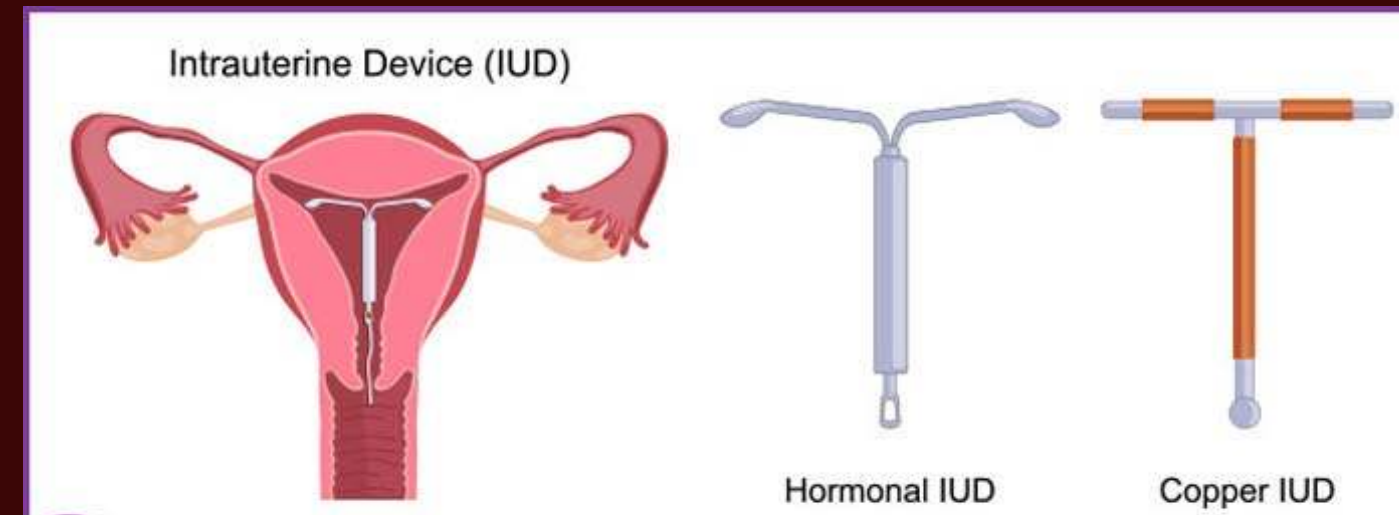
2. Chemical Methods:

Hormonal contraceptives, such as oral and vaginal pills, prevent ovulation, alter cervical mucus, or change the uterine lining to prevent implantation. However, these methods may have side effects and do not protect against sexually transmitted diseases (STDs).



3. Intrauterine Devices (IUDs):

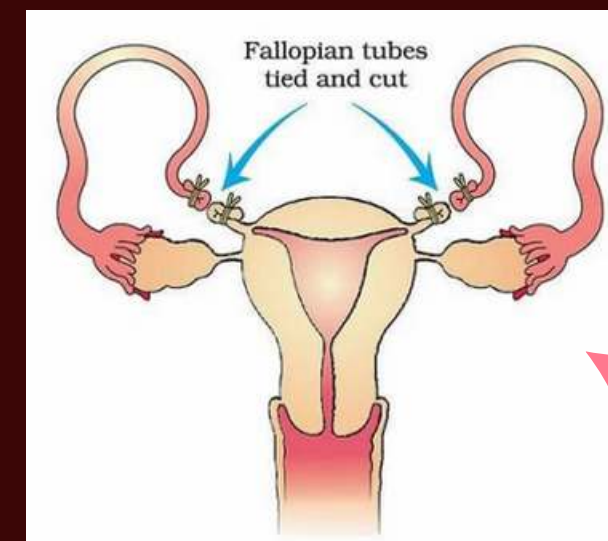
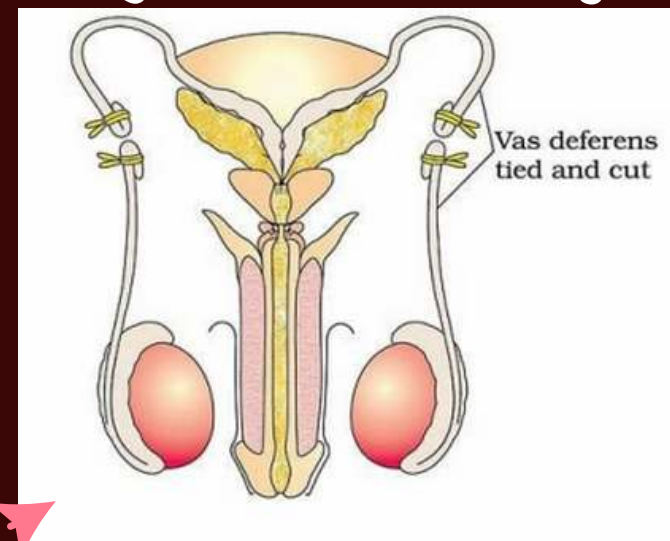
- These are small T-shaped devices that are planted in the uterus.
- Copper-T or Loop placed in the uterus to prevent pregnancy.
- Can cause uterine irritation in some cases.



4. Surgical Methods (Permanent):

- **Male (Vasectomy):** Blocking the vas deferens to prevent sperm transfer.
- **Female (Tubectomy):** Blocking the fallopian tubes to prevent egg transfer.
- Surgery is safe in the long run but may have risks like infection.

Vasectomy



Tubectomy

Q. Which of the following is a surgical method of contraception?

- (A) Use of a condom**
- (B) Copper-T insertion**
- (C) Vasectomy**
- (D) Oral pills**

Q. Which among the following diseases is not sexually transmitted?

- a) Syphilis**
- b) Hepatitis**
- c) HIV-AIDS**
- d) Gonorrhea**

Q. Suggest three contraceptive methods to control the size of human population which is essential for the health and prosperity of a country. State the basic principle involved in each.

Q. Suggest three contraceptive methods to control the size of human population which is essential for the health and prosperity of a country. State the basic principle involved in each.

Answer: Three contraceptive methods which can help to control human population are:

(i) Condom : It is a mechanical barrier which does not allow sperms and ovum to meet, hence prevents fertilisation. Condoms are made of thin rubber/latex sheath used to cover the penis in the male and vagina/cervix in female just before coitus (intercourse) so that the ejaculated semen is not released in the female reproductive tract.

(ii) Intrauterine devices (IUDs): These are devices inserted by doctors or expert nurses in the uterus through vagina. These are presently available as non-medicated IUDs, copper releasing IUDs (CuT, etc.) and hormone releasing IUDs. They increase phagocytosis of sperms within uterus and suppress sperm motility and its fertilising capacity. They also make uterus unsuitable for implantation and cervix hostile to sperms.

(iii) Oral pills : These pills contain progesterone alone or a combination of progestogen and estrogen. They inhibit ovulation and make uterus unsuitable for implantation, hence prevent fertilisation.

Q: Name the part/organ of the human female reproductive system

- 1. where contraceptive devices such as loop or copper-T are placed to prevent pregnancy.**
- 2. which is blocked to prevent the transfer of eggs.**
- 3. where formation of green cells as ova takes place.**
- 4. from where the embryo gets nutrition from the mother's blood.**

Q: Name the part/organ of the human female reproductive system (2022)

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- 2. which is blocked to prevent the transfer of eggs.**
- 3. where formation of green cells as ova takes place.**
- 4. from where the embryo gets nutrition from the mother's blood.**

Ans:

- 1. The part/organ of the human female reproductive system where contraceptive devices such as loop or copper-T are placed to prevent pregnancy is the uterus or womb. These devices are inserted into the uterus to inhibit fertilization and implantation of a fertilized egg.**
- 2. The part/organ of the human female reproductive system that is blocked to prevent the transfer of eggs is the fallopian tubes. These tubes connect the ovaries to the uterus, and they are blocked or sealed off in certain contraceptive methods or sterilization procedures to prevent the eggs from reaching the uterus for fertilization.**
- 3. The part/organ of the human female reproductive system where the formation of green cells as ova takes place is the ovaries. Ovaries are responsible for producing and releasing mature eggs, or ova, during the menstrual cycle.**
- 4. The part/organ of the human female reproductive system from where the embryo gets nutrition from the mother's blood is the placenta. The placenta is a temporary organ that develops during pregnancy and attaches to the uterine wall. It facilitates the exchange of nutrients, oxygen, and waste products between the mother's blood and the developing fetus.**

Q. The growing size of the human population is a cause of concern for all people. The rate of birth and death in a given population will determine its size. Reproduction is the process by which organisms increase their population. The process of sexual maturation for reproduction is gradual and takes place while general body growth is still going on. Some degree of sexual maturation does not necessarily mean that the mind or body is ready for sexual acts or for having and bringing up children. Various contraceptive devices are being used by human beings to control the size of population.

(i) List two common signs of sexual maturation in boys and girls. (ii) What is the result of reckless female Foeticide?

(iii) Which contraceptive method changes the hormonal balance of the body? (iv) Write two factors that determine the size of a population. (2020)

Q. The growing size of the human population is a cause of concern for all people. The rate of birth and death in a given population will determine its size. Reproduction is the process by which organisms increase their population. The process of sexual maturation for reproduction is gradual and takes place while general body growth is still going on. Some degree of sexual maturation does not necessarily mean that the mind or body is ready for sexual acts or for having and bringing up children. Various contraceptive devices are being used by human beings to control the size of population.

(i) List two common signs of sexual maturation in boys and girls. (ii) What is the result of reckless female Foeticide?

(iii) Which contraceptive method changes the hormonal balance of the body? (iv) Write two factors that determine the size of a population.

Ans: i Two common signs of sexual maturation in boys and girls are:

1. Growth of pubic hair and extra hair in the armpits.

2. Development of oily skin and pimples.

(ii) Female Foeticide is reducing the number of girls drastically in our country, due to which male-female sex ratio is also declining.

(iii) Chemical contraceptive method changes the hormonal balance of the body.

(iv) The rate of birth and death in a given population will determine the size of a population.

Class 10th Phodenge!!

